

**Gradient Conventionalization: A Unified Framework for Lexical Semantics and Social
Epistemology**

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Abstract

This paper proposes a unified theoretical framework that reconceptualizes the relationship between lexical meaning and world knowledge as a gradient of conventionalization rather than a categorical distinction. Through analysis of entailment strength and intersubjective agreement patterns, I argue that meaning components should be distinguished by their level of conventionalization—operationalized as degrees of speaker agreement—rather than by categorical membership in "lexical" versus "encyclopedic" knowledge. The framework introduces a five-level gradient (reference, denotation, sense, cultural connotation, private connotation) corresponding to decreasing levels of intersubjective agreement, and develops a typology of entailments (strong, default, weak) correlated with these conventionalization levels. This reconceptualization resolves longstanding puzzles in semantic prosody, explains why certain types of arguments converge while others polarize, and provides a unified account connecting lexical semantics to social epistemology. The framework generates testable predictions about semantic change, cross-linguistic variation, language acquisition, and the dynamics of disagreement in scientific versus philosophical discourse.

Keywords: lexical semantics, conventionalization, semantic prosody, entailment, intersubjective agreement, epistemology of disagreement

Gradient Conventionalization: A Unified Framework for Lexical Semantics and Social Epistemology

The relationship between what words mean and what speakers know presents one of the most persistent theoretical challenges in semantics. On one hand, formal semantic theory has developed sophisticated frameworks for analyzing compositional meaning, truth conditions, and logical entailments, treating lexical knowledge as systematic, categorical, and shared across speakers. On the other hand, corpus linguistics has documented systematic patterns in word usage that resist traditional semantic analysis—patterns that show statistical regularity yet remain largely inaccessible to conscious introspection. This tension is perhaps nowhere more evident than in semantic prosody research, where words like *cause*, *commit*, and *regime* exhibit consistent distributional skews toward particular evaluative contexts despite speakers' inability to consciously articulate these patterns (Louw, 1993; Sinclair, 1991; Stubbs, 2001).

Traditional approaches to this puzzle have relied on a categorical distinction between lexical (or linguistic) knowledge and encyclopedic (or world) knowledge. Under this view, lexical semantic properties constitute the proper domain of linguistic theory—systematic, compositional, shared across speakers—while encyclopedic associations belong to pragmatics, cultural knowledge, or individual psychology (Katz & Fodor, 1963; Chomsky, 1965). This boundary has been invoked to explain why certain patterns are theorizable while others remain resistant to formalization. However, the boundary itself has proven remarkably unstable, with theorists disagreeing about which aspects of meaning properly belong on which side (Haiman, 1980; Jackendoff, 2002; Langacker, 1987).

I propose that this instability reflects a fundamental error in conceptualization: the lexical-encyclopedic distinction is not a categorical boundary between types of knowledge but rather a gradient of conventionalization. Both linguistic meaning and world knowledge, I argue,

are ultimately encyclopedic in nature—that is, learned, arbitrary, and contingent—with what we call "lexical" knowledge representing simply the highest degree of intersubjective agreement and conventionalization. This reconceptualization has profound implications for understanding connotative meaning, which has remained chronically undertheorized in formal semantics despite its obvious systematicity in corpus patterns.

The key diagnostic error in prior work has been the conflation of two distinct phenomena under the single label "connotation": *cultural connotation* (systematic, community-level patterns with moderate intersubjective agreement) and *private connotation* (idiosyncratic, individual associations with minimal agreement). Cultural connotation generates what I term *weak entailments*—systematic but defeasible inferences that pattern differently from classical logical entailments but nevertheless show predictable cross-speaker consistency. Private connotation, by contrast, consists of purely associative networks that vary across individuals and lack the systematicity required for theoretical treatment at the community level.

By failing to distinguish these phenomena, semantic theory has treated all connotation as if it were private, thereby concluding that connotative meaning lies outside the proper scope of semantic analysis. This conclusion is premature and empirically incorrect. Cultural connotation exhibits sufficient systematicity to warrant formal treatment, as evidenced by consistent corpus patterns, cross-speaker agreement on subtle semantic judgments, and the successful extraction of such patterns by computational language models.

Beyond resolving specific empirical puzzles in lexical semantics, this framework provides theoretical tools for understanding a seemingly unrelated phenomenon: the epistemology of disagreement. The same gradient of conventionalization that distinguishes types of lexical meaning also predicts patterns in argumentative discourse. When interlocutors argue about matters involving high-agreement meanings (reference, denotation), they can typically

reach consensus through evidence and reasoning. When they argue about matters involving moderate-agreement meanings (sense), arguments become unproductive—participants talk past each other without convincing. When they argue about matters involving low-agreement meanings (cultural and private connotation), arguments actively widen differences and increase polarization.

This connection between semantic structure and argument dynamics explains several puzzling asymmetries: why scientific communities converge on theoretical consensus while philosophical communities do not; why political discourse has become increasingly polarized as it has shifted toward arguing about contested terms; why some metalinguistic negotiations succeed while others fail. The gradient framework thus unifies lexical semantics and social epistemology, demonstrating that the structure of meaning constrains the dynamics of belief revision and consensus formation.

This paper develops these arguments in seven sections. Section 2 examines the traditional lexical-encyclopedic dichotomy and presents five converging lines of evidence that this boundary is better understood as a gradient than a categorical distinction. Section 3 develops a formal framework for understanding meaning as graded conventionalization, operationalized through levels of intersubjective agreement. Section 4 introduces the distinction between strong, default, and weak entailments, arguing that entailment strength correlates with conventionalization level. Section 5 diagnoses the conflation of cultural and private connotation and demonstrates how this has led to the persistent undertheorization of connotative meaning. Section 6 applies the framework to semantic prosody phenomena, showing how treating prosodic patterns as cultural connotation generating weak entailments resolves several longstanding puzzles. Section 7 explores epistemological applications, demonstrating how the gradient framework predicts and explains patterns in argumentative discourse and disagreement

dynamics. Section 8 examines broader implications for semantic theory, language acquisition, computational modeling, and cross-linguistic semantics, and Section 9 concludes by outlining future research directions.

The Instability of the Lexical-Encyclopedic Boundary

The distinction between lexical (linguistic, semantic) knowledge and encyclopedic (world, conceptual) knowledge has played a foundational role in formal semantics since Katz and Fodor's (1963) pioneering work on semantic theory. The basic intuition is straightforward: linguistic knowledge comprises those aspects of meaning that are properly part of the language system—systematic, compositional, shared across speakers—while encyclopedic knowledge consists of facts about the world that speakers happen to know. Under this view, knowing that *bachelor* means "unmarried adult male" constitutes linguistic knowledge, while knowing that bachelors typically live in apartments or eat takeout food represents encyclopedic knowledge.

This distinction has been invoked to solve several theoretical problems. It provides a principled basis for limiting the scope of semantic theory, preventing the incorporation of all world knowledge into lexical entries. It offers an explanation for why certain inferences pattern like entailments while others seem more like defeasible expectations. And it aligns with intuitions about compositionality: linguistic meanings combine systematically to generate complex meanings, while encyclopedic knowledge does not obviously compose in the same way (Fodor & Lepore, 1992).

However, despite its theoretical utility, the lexical-encyclopedic boundary has proven remarkably difficult to maintain in practice. Attempts to draw a clear line have consistently encountered problems, with theorists disagreeing about the proper classification of numerous meaning components. Consider the following cases: Is it linguistic knowledge that *kill* means "cause to die"? Most would say yes. But is it linguistic knowledge that killing typically involves

force or violence? That killing is generally considered morally wrong? That accidental killings differ morally from intentional ones? Different theories give different answers, and the boundary becomes increasingly unclear as we move from core definitional features to more peripheral but still systematic associations.

Similar problems arise with color terms. Knowing that *red* refers to a particular region of the color spectrum seems clearly linguistic. But what about knowing that red is associated with danger, passion, or heat in many cultures? These associations show remarkable consistency across speakers and influence actual language use in systematic ways, yet they are typically classified as encyclopedic rather than linguistic. The problem is not merely one of borderline cases. Rather, sustained investigation suggests that the boundary may not be a natural one.

Theoretical Disagreements Across Frameworks

The lexical-encyclopedic boundary is persistently contested across different theoretical frameworks. Generative semanticists in the 1970s argued for incorporating more world knowledge into semantic representations (Lakoff, 1971), while formal semanticists sought to maintain a stricter boundary (Katz, 1977). Cognitive linguists have largely rejected the distinction altogether, arguing that all meaning is essentially encyclopedic (Croft & Cruse, 2004; Langacker, 1987). Frame semantics acknowledges that understanding even basic word meanings requires access to structured background knowledge (Fillmore, 1982).

Fillmore's (1982) analysis of *bachelor* famously demonstrated that this supposedly paradigmatic case of lexical definition actually requires extensive cultural and social knowledge for appropriate application. The word cannot be adequately defined through necessary and sufficient conditions precisely because its application depends on culturally specific knowledge about marriage practices, typical life stages, and social expectations. A Catholic priest, though unmarried and adult male, is not typically called a bachelor; nor is the Pope, despite satisfying

the definitional criteria. These facts suggest that even apparently core lexical items cannot be cleanly separated from encyclopedic knowledge.

These persistent theoretical disagreements suggest that the boundary may not be a robust empirical phenomenon but rather a theoretical construct whose utility varies across frameworks. If the distinction reflected a natural division in the structure of meaning, we would expect greater theoretical consensus about where to draw it. Instead, different frameworks locate it at radically different positions, or reject it entirely.

Psycholinguistic Processing Evidence

Psycholinguistic evidence fails to support a categorical processing distinction between lexical and encyclopedic knowledge. If these were truly distinct types of knowledge, we might expect to see different processing signatures, separate neural substrates, or differential vulnerability to brain damage. While there is evidence for some specialization in language processing, the picture that emerges is one of graded differences and extensive interaction rather than categorical separation (Federmeier & Kutas, 1999; Hagoort et al., 2004).

Semantic priming studies show that both "core" semantic features and "peripheral" encyclopedic associations can facilitate processing, with the strength of priming effects varying continuously rather than categorically (McRae et al., 2005). For instance, the word *dog* primes both *animal* (a definitional feature) and *bone* (an encyclopedic association), with priming strength reflecting factors like frequency and typicality rather than categorical membership in lexical versus encyclopedic knowledge. If the distinction were psychologically real, we would expect qualitatively different priming patterns or processing timecourses for the two types of information.

Event-related potential (ERP) studies similarly fail to reveal categorical differences. The N400 component, which indexes semantic processing difficulty, shows graded sensitivity to

semantic violations regardless of whether those violations involve "core" lexical features or "peripheral" world knowledge (Kutas & Federmeier, 2011). Violations of definitional features (e.g., "The bachelor was married") and violations of typical associations (e.g., "The bachelor attended the quilting circle") both elicit N400 effects, with amplitude differences reflecting gradations in expectation violation rather than categorical type differences.

Neuroimaging studies have sought to identify distinct neural substrates for linguistic versus conceptual knowledge, but results have been mixed at best. While certain brain regions show preferential involvement in language processing, there is extensive overlap in the neural systems supporting lexical semantic knowledge and general conceptual knowledge (Binder et al., 2009). Attempts to find "language-specific" semantic regions distinct from general conceptual processing have largely failed, suggesting that lexical and encyclopedic knowledge may be neurally integrated rather than categorically separated.

Cross-Linguistic Variation Patterns

The lexical-encyclopedic distinction becomes particularly problematic when examining cross-linguistic variation. Languages differ not only in which concepts they lexicalize but also in how they organize semantic space. The classic example involves color terms: while all languages make color distinctions, the specific boundaries and number of basic color categories vary systematically across languages (Berlin & Kay, 1969). If color category boundaries were part of universal conceptual structure (encyclopedic), we would expect more uniformity. If they were purely linguistic, we would expect more arbitrary variation. The actual pattern—systematic variation constrained by perceptual and cognitive factors—suggests a middle ground that the categorical distinction fails to capture.

More generally, what counts as "lexical" meaning in one language often corresponds to pragmatic inference or encyclopedic knowledge in another. Many languages grammatically

encode evidentiality—the source of information underlying an assertion—through obligatory morphological marking (Aikhenvald, 2004). In these languages, evidential distinctions clearly constitute part of the grammatical system and thus linguistic knowledge. In English, similar distinctions are expressed through optional adverbials (*apparently, reportedly*) or pragmatic inference, and thus might be classified as encyclopedic. But the semantic content—the distinction between direct versus indirect evidence—is the same. The cross-linguistic variation demonstrates that the boundary between lexical and encyclopedic is language-specific rather than reflecting universal semantic architecture.

Similarly, kinship terminology systems vary dramatically across languages in which distinctions are lexicalized versus inferred from context (Kroeber, 1909). English lexicalizes the distinction between *grandfather* (father's father) and *grandfather* (mother's father) only through modification, while some languages have distinct basic terms. Does this mean speakers of those languages have different "lexical" knowledge than English speakers? Or does it simply mean that the same conceptual distinctions are conventionalized at different levels? The cross-linguistic evidence suggests the latter—that conventionalization level varies across languages, rather than there being a universal partition of knowledge into lexical and encyclopedic types.

Language Acquisition Mechanisms

Language acquisition patterns do not support learning mechanisms that would differentially acquire lexical versus encyclopedic knowledge. Children learn word meanings through the same evidence that they learn facts about the world: observing language use in context and extracting statistical patterns (Tomasello, 2003). They do not appear to have separate learning mechanisms for acquiring "linguistic" meanings versus "world" knowledge. The gradual acquisition of word meanings, with increasing precision and refinement over time, suggests a

continuous process of hypothesis formation and revision rather than the discovery of discrete categorical boundaries (Clark, 2009).

Consider children's acquisition of *bachelor*. Initially, children may understand only that the word refers to unmarried males. Over time, they refine this understanding to include age restrictions (adults, not children), social expectations (eligibility for marriage), and contextual appropriateness (not applicable to priests or committed long-term partners). This refinement process does not involve first acquiring a "lexical" core and then adding "encyclopedic" periphery; rather, it involves continuous elaboration of a unified semantic representation that integrates linguistic convention with cultural knowledge (Carey, 2009).

Developmental studies show that children's semantic development follows similar trajectories for features that would be classified as "lexical" versus "encyclopedic." Both types of information become more precise and adult-like gradually, through the same learning mechanisms operating on the same input evidence. If the distinction were psychologically real, we might expect different developmental timecourses, but this is not what we observe (Landau & Gleitman, 1985).

Computational Language Models

The success of computational language models trained purely on distributional patterns challenges the necessity of the lexical-encyclopedic distinction. Large language models acquire representations that capture both what traditional theory calls "lexical" meaning and what it calls "encyclopedic" knowledge, despite making no architectural distinction between these types (Brown et al., 2020; Devlin et al., 2019). These models learn from next-word prediction tasks on massive text corpora, receiving no explicit instruction distinguishing "linguistic" from "world" information.

Yet these systems develop both syntactic and semantic competence typically attributed to linguistic knowledge (correctly conjugating irregular verbs, observing subcategorization restrictions, demonstrating sensitivity to semantic anomaly) and extensive factual knowledge about the world typically classified as encyclopedic (answering factual questions, reasoning about causal relationships, demonstrating domain knowledge across disciplines). Critically, they acquire these different knowledge types through identical mechanisms operating on the same distributional evidence.

If lexical and encyclopedic knowledge were categorically distinct types requiring different learning mechanisms or different training signals, language models' success would be mysterious. Instead, their performance suggests that the distributional patterns in language use encode both types of information in a unified way, making both recoverable through the same statistical learning procedures (Bender & Koller, 2020). The fact that such models achieve human-like performance on semantic tasks without respecting the lexical-encyclopedic boundary provides strong evidence that this boundary may not be necessary for representing or processing meaning.

Recent probing studies have examined what semantic information is encoded in language model representations (Petroni et al., 2019; Ettinger, 2020). These studies find that models develop representations sensitive to both "core" definitional features and "peripheral" world knowledge, with no clear categorical separation between the two. Instead, different types of semantic information appear to be continuously distributed across the representational space, with distinctions of degree rather than kind.

Converging Implications

These five converging lines of evidence—theoretical instability, psycholinguistic continuity, cross-linguistic variation, unified acquisition, and computational modeling success—

collectively suggest that the lexical-encyclopedic boundary is not a natural division in the structure of meaning but rather an artifact of particular theoretical commitments. If the boundary is not natural, we need an alternative framework for understanding how different aspects of meaning relate to one another and why some aspects seem more "core" or "linguistic" than others.

The alternative I propose treats meaning as existing on a continuous gradient of conventionalization, where what distinguishes different aspects of meaning is not their categorical membership in distinct knowledge types but rather their level of intersubjective agreement among speakers. This reconceptualization preserves the insight that some knowledge is more stable, more widely shared, and more central to linguistic competence, while avoiding the problems that arise from attempting to draw categorical boundaries through continuous space.

Meaning as Graded Conventionalization

I propose that what distinguishes different aspects of meaning is not their membership in categorically distinct types of knowledge but rather their level of conventionalization, operationalized as the degree of intersubjective agreement among speakers. Intersubjective agreement refers to the extent to which members of a speech community share the same associations, inferences, or judgments about a word's meaning. This reconceptualization treats meaning as existing on a continuous gradient rather than in discrete categories.

The Foundation: Arbitrariness of Linguistic Signs

The foundation of this proposal rests on a fundamental property of linguistic signs: their arbitrariness. As Saussure (1916/1959) emphasized, the relationship between a word and its meaning is conventional rather than natural. There is no intrinsic reason why the sound sequence /dɒg/ should refer to canines in English; this association is purely a matter of social convention.

Importantly, this arbitrariness extends beyond the sound-meaning pairing to all aspects of semantic structure.

The fact that English *bachelor* encodes both "unmarried" and "adult male" as definitional features while leaving marital desirability or apartment-dwelling as peripheral is itself a matter of convention. Different languages make different choices about which features to encode lexically and which to leave to inference. The arbitrariness principle thus applies not only to the Saussurean sign (the pairing of signifier and signified) but to the internal structure of the signified itself—to which semantic features are conventionally associated with a given concept at which levels of entrenchment.

If all aspects of word meaning are ultimately conventional—learned through exposure to language use in specific cultural and social contexts—then what we call "linguistic" or "lexical" knowledge simply represents the highest levels of conventionalization: those aspects of meaning that show near-universal agreement among speakers of a language community. Conversely, what we call "encyclopedic" or "world" knowledge represents lower levels of conventionalization: aspects of meaning that show more variation across speakers, subgroups, or contexts.

This gradient view is consistent with usage-based approaches to language that emphasize the role of frequency and entrenchment in shaping linguistic representations (Bybee, 2006; Goldberg, 2006). However, it differs from some usage-based accounts by proposing a continuous gradient rather than multiple discrete levels, and by operationalizing the gradient through intersubjective agreement rather than individual frequency of exposure.

Formal Framework for Gradient Conventionalization

Let $\mathcal{S}$ denote a speech community and w a word in the shared language $\mathcal{L}$. Let $\mathcal{F}$ represent the space of all possible semantic features or associations that speakers might have with w . For

any feature $f \in \mathcal{F}$, we can define an agreement function $\alpha: \mathcal{F} \rightarrow [0,1]$ that measures the proportion of speakers in $\mathcal{S}$ who associate feature f with word w .

More formally, let $S_f = \{s \in \mathcal{S}: s \text{ associates } f \text{ with } w\}$ be the set of speakers who make the association between feature f and word w . Then:

$$\alpha(f) = \frac{|S_f|}{|\mathcal{S}|}$$

This agreement score $\alpha(f)$ ranges from 0 (no speakers make the association) to 1 (all speakers make the association). In practice, $\alpha(f)$ can be operationalized through multiple converging measures:

1. **Explicit judgment tasks:** Speakers rate the strength of association between features and words on Likert scales, and agreement is computed as the proportion giving ratings above a threshold.
2. **Corpus-based distributional consistency:** Agreement is estimated from the consistency of distributional patterns across diverse text types and registers, under the assumption that higher agreement produces more consistent usage.
3. **Experimental priming:** Agreement is inferred from the strength and consistency of priming effects across participants, with stronger and more consistent priming indicating higher agreement.
4. **Computational measures:** Agreement is estimated from the activation strength of features in language models trained on naturalistic data from multiple sources, under the assumption that higher agreement produces stronger learned associations.

The convergence across these measures provides robust evidence for conventionalization levels. Importantly, while I present α as a single-valued function for simplicity, in practice it may

be multidimensional, with different aspects of agreement (e.g., judgment consensus, behavioral consistency, distributional regularity) potentially dissociating.

The Five-Level Gradient

While conventionalization is fundamentally continuous, we can identify several reference points along this gradient that correspond to theoretically and empirically meaningful distinctions. Let n be a constant representing the characteristic "step size" between conventionalization levels. I propose five reference points, defined by their agreement scores:

Level 1: Reference ($\alpha \approx 1.0$, or $\alpha = x$ where $x \rightarrow 1$)

Reference comprises the basic extensional mapping that defines which entities in the world a word applies to. This represents near-universal agreement on the word's extension. For instance, virtually all English speakers agree that *dog* applies to four-legged canine mammals and not to cats, chairs, or abstract concepts. Disagreements at this level are vanishingly rare and typically involve genuine borderline cases (Is a wolf a dog? Is a fox?) where the disagreement reflects genuine vagueness in category boundaries rather than variation in conventionalization.

Formally, if f_{ref} denotes the reference feature (the mapping from word to extension), then $\alpha(f_{\text{ref}}) = x$ where x is very close to 1. Deviations from perfect agreement ($x < 1$) arise primarily from measurement error, borderline cases, and the small proportion of speakers with idiosyncratic or deficient lexical knowledge.

Level 2: Denotation ($\alpha = x - n$, where n is small)

Denotation comprises core definitional features showing very high but not quite universal agreement. These are the features typically identified in dictionary definitions as necessary conditions for application of the word. For *bachelor*, the features "unmarried," "adult," and "male" all fall at this level. The inference from " x is a bachelor" to " x is unmarried" is made by the vast majority of speakers, with agreement scores typically above 0.90.

The slight decrement from perfect reference-level agreement arises because denotational features, while highly conventionalized, can sometimes be overridden in extended or metaphorical uses, or may be subject to minor cross-speaker variation in how strictly they are applied. Some speakers might accept calling a long-term committed but unmarried partner a "bachelor" in certain contexts, creating small deviations from universal agreement.

Level 3: Sense ($\alpha = x - 2n$)

Sense comprises features defining how a word relates to other words in its semantic field—the network of contrasts, similarities, and structural relationships that position the word within the larger semantic system. For *bachelor*, sense-level features include its contrast with *married man*, its alignment with *eligible*, and its position within the social-status dimension of male identity terms.

Sense-level features typically show agreement in the range of 0.70-0.90. Most speakers recognize these relationships and use them to structure semantic space, but there is more variation than at the denotation level. Some speakers may organize semantic fields differently, prioritize different contrasts, or have different intuitions about near-synonymy and antonymy relationships.

Level 4: Cultural Connotation ($\alpha = x - 3n$)

Cultural connotation comprises systematic evaluative or stylistic associations showing moderate agreement within a speech community. These are the associations that corpus linguists identify as semantic prosodies—the tendency for words to appear in contexts with particular evaluative or affective characteristics. For *regime*, the association with authoritarian or oppressive governance falls at this level. For *cause*, the association with negative outcomes represents cultural connotation.

Agreement scores at this level typically range from 0.50-0.70. A majority of speakers share these associations and they produce systematic corpus patterns, but there is substantial variation. Some speakers may lack the association entirely, others may have it weakly, and still others may have it strongly. This variation reflects differences in language exposure, register familiarity, and individual sensitivity to distributional patterns.

Level 5: Private Connotation ($\alpha = x - 4n$, where $\alpha < 0.50$)

Private connotation comprises idiosyncratic associations arising from individual experience, showing minimal cross-speaker agreement. These are the personal, autobiographical associations that individuals develop through their unique life experiences. For instance, *dog* might evoke childhood memories of a beloved pet for one speaker, memories of a traumatic dog bite for another, and neutral or no particular association for a third.

Agreement scores at this level fall below 0.50, often approaching zero for highly specific personal associations. These associations are not systematic at the community level and do not produce predictable corpus patterns. They are psychologically real for individuals but lack the intersubjective stability required for conventional meaning.

Mathematical Representation

We can represent the overall semantic content of word *w* as a weighted distribution over features, where the weight reflects conventionalization level:

$$\text{Sem}(w) = \sum_{f \in \mathcal{F}} \alpha(f) \cdot f$$

This representation captures the intuition that a word's meaning is not a discrete set of features but rather a graded distribution, with some features more central (higher α) and others more peripheral (lower α). Different theoretical frameworks can be understood as implicitly

setting different thresholds θ such that features with $\alpha(f) > \theta$ are considered "lexical" while those with $\alpha(f) \leq \theta$ are considered "encyclopedic."

The gradient view maintains that no such threshold is theoretically privileged—the distinction is one of degree throughout. However, for practical purposes, we might define:

- **Core lexical features:** $\alpha(f) > 0.90$ (reference and denotation)
- **Extended lexical features:** $0.70 < \alpha(f) \leq 0.90$ (sense)
- **Systematic conventional features:** $0.50 < \alpha(f) \leq 0.70$ (cultural connotation)
- **Idiosyncratic features:** $\alpha(f) \leq 0.50$ (private connotation)

Properties of the Gradient Framework

Several properties of this framework warrant emphasis:

Continuity: The gradient is fundamentally continuous rather than consisting of discrete levels. The five-level partition above represents convenient reference points for theoretical discussion rather than natural boundaries. Empirically, we should expect to find features at all intermediate agreement levels, not clustering at discrete points.

Within-word variation: Different meaning components of the same word may occupy different positions on the gradient. While the basic denotation of *bachelor* (unmarried adult male) shows very high agreement, its cultural connotations (associations with particular lifestyles, desirability, social positions) show much more variation across speakers and communities.

Cross-linguistic variation: The conventionalization level of parallel semantic features can vary across languages. What is highly conventionalized in one language (high α) may be less conventionalized in another (lower α), even when the languages express the same conceptual content.

Diachronic change: Conventionalization levels can change over time through semantic change processes. Features that begin with low agreement (cultural or private connotation) can become more conventionalized (moving toward sense or denotation), and vice versa through processes of semantic bleaching or weakening.

Individual variation: While α represents population-level agreement, individual speakers may deviate from the population average. A speaker might have stronger or weaker associations for particular features than the community norm, representing individual variation around the central tendency.

Empirical Predictions

The gradient framework generates several testable empirical predictions:

P1: Continuous variation: If meaning is gradient rather than categorical, agreement scores should vary continuously rather than clustering at discrete points. Corpus studies measuring distributional consistency and experimental studies measuring feature verification times should reveal continuous distributions rather than bimodal or multimodal clustering.

P2: Agreement-processing correlation: Agreement level should correlate with processing measures. Features with higher α should show faster verification times, stronger priming effects, and reduced N400 amplitudes when violated. This correlation should be continuous rather than showing discrete jumps at category boundaries.

P3: Acquisition gradient: Language acquisition should show gradient patterns corresponding to conventionalization levels. Higher-agreement features (α closer to 1) should be learned earlier and more robustly than lower-agreement features. Developmental trajectories should be continuous rather than showing discrete stages corresponding to categorical knowledge types.

P4: Cross-linguistic systematicity: For translation-equivalent words across languages, the rank ordering of features by agreement level should show cross-linguistic consistency, even if absolute agreement levels vary. Features that are highly conventionalized in one language should tend to be relatively more conventionalized in translation equivalents.

P5: Individual-demographic correlation: Individual variation in agreement scores should pattern with demographic and experiential factors. Speakers with more exposure to particular registers or domains should show higher agreement on features conventional in those contexts.

Preliminary evidence supports several of these predictions. Psycholinguistic studies using feature verification tasks show continuous variation in response times and error rates across different semantic features, rather than a categorical split between lexical and encyclopedic (McRae et al., 2005). Developmental studies document gradual refinement of word meanings rather than sudden acquisition of categorical boundaries (Carey, 2009). Cross-linguistic research demonstrates that features encoded as "core" meaning in one language may be peripheral or pragmatic in another, but with systematic cross-linguistic correlations (Malt et al., 1999).

However, comprehensive empirical testing of these predictions remains a priority for future research. The framework developed here provides the theoretical foundation for such testing by specifying precisely what should be measured (intersubjective agreement across multiple operationalizations) and what patterns should emerge (continuous gradients, systematic correlations, predictable individual variation).

A Theory of Weak Entailments

The reconceptualization of meaning as graded conventionalization requires a parallel reconceptualization of how meanings generate inferences. Traditional semantic theory focuses primarily on strong entailments: inferences that follow necessarily from meaning and cannot be

cancelled without contradiction. However, if meaning exists on a gradient, we should expect entailments to exist on a corresponding gradient of strength. This section develops a typology of entailments distinguished by their conventionalization levels and cancellability properties.

Beyond Classical Entailment

An entailment, in the classical sense, is a relation between propositions such that whenever the first proposition is true, the second must also be true. For example, "John is a bachelor" entails "John is unmarried"; it is impossible for the first to be true while the second is false (assuming standard interpretation). This notion of entailment plays a foundational role in formal semantics, serving as a diagnostic for semantic content: if a proposition p entails a proposition q , then the information expressed by q is part of the meaning of p (Chierchia & McConnell-Ginet, 2000).

Formally, we can define classical entailment as:

$$p \models q \text{ iff } \forall w \in W: w \models p \implies w \models q$$

where W is the set of possible worlds and $w \models p$ means proposition p is true in world w . Under this definition, entailment is categorical: either p entails q or it does not. There is no intermediate ground.

However, many systematic patterns in language use do not fit this classical pattern. Consider the verb *cause*. Corpus analysis reveals that *cause* appears predominantly in contexts describing negative or undesirable outcomes (Stubbs, 2001). Yet it is perfectly acceptable to say "The policy caused economic growth" or "The treatment caused a full recovery." The association with negativity is systematic and influences speakers' expectations, but it does not constitute a classical entailment because it can be overridden without contradiction.

Traditional approaches handle such cases by relegating them to pragmatics: the negative association is not part of *cause*'s semantic meaning but rather a conversational implicature or statistical generalization about usage patterns (Grice, 1975). However, this move fails to capture the systematicity of the pattern. Pragmatic implicatures should be highly context-dependent and easily cancelled, yet the prosodic patterns associated with words like *cause* are remarkably stable across corpora and contexts. We need a richer typology of entailments that recognizes gradations in entailment strength corresponding to gradations in conventionalization.

Three Types of Entailments

I propose a three-way typology of entailments corresponding to different regions of the conventionalization gradient:

Strong Entailments: Arising from reference and denotation levels ($\alpha \geq x - n$)

Strong entailments correspond to the classical notion: inferences that follow necessarily from meaning and cannot be cancelled without contradiction. These arise from the highest levels of conventionalization, where speaker agreement approaches universality.

Examples include:

- *bachelor* → unmarried
- *kill* → cause to die
- *dog* → animal
- *triangle* → three-sided polygon

Strong entailments are characterized by several properties:

1. *Near-universal speaker agreement:* Virtually all competent speakers make the inference.
2. *Stability across contexts:* The entailment holds regardless of discourse context, genre, or register.
3. *Resistance to cancellation:* Denying the entailment results in contradiction or zeugma.

4. *Non-defeasibility*: Additional information cannot override the entailment without forcing reinterpretation.

Formally, we can represent strong entailments as:

$$\text{If } \alpha(f) \geq x - n, \text{ then } p(w) \models f \text{ and } \neg \diamond (\neg f \mid p(w))$$

where $p(w)$ is a proposition asserting property w of some entity, f is the entailed feature, and $\neg \diamond (\neg f \mid p(w))$ indicates that it is not possible to deny f while maintaining $p(w)$ without contradiction.

The cancellation test definitively distinguishes strong entailments: "John is a bachelor but he is married" is contradictory unless *bachelor* is being used in a different sense (such as "bachelor's degree"). The semantic content that must be preserved for the word to retain its standard meaning constitutes the strong entailment.

Default Entailments: Arising from sense level ($\alpha \approx x - 2n$)

Default entailments are systematic inferences that normally follow from meaning but can be overridden by explicit contextual information. These arise from high but not maximal conventionalization, corresponding to the sense level of the gradient.

Examples include:

- *bird* $\rightarrow$ can fly
- *open* (door) $\rightarrow$ create opening by moving door on hinges
- *mother* $\rightarrow$ biological female parent (vs. adoptive mother)
- *see* $\rightarrow$ perceive visually with functioning eyes (vs. blind person who "sees" tactilely)

Default entailments show the following properties:

1. *High speaker agreement*: Most speakers make the inference in neutral contexts.
2. *Context sensitivity*: The strength of the inference varies with contextual support.

3. *Cancellability without contradiction*: Denying the entailment is acceptable when explicitly marked.
4. *Processing effects*: The default inference is activated even when ultimately overridden.

Formally, default entailments can be represented using defeasible logic:

$$\text{If } \alpha(f) \approx x - 2n, \text{ then } p(w) \Rightarrow f \text{ and } \diamond (\neg f \mid p(w) \wedge \text{Context})$$

where $\Rightarrow$ represents defeasible entailment and $\diamond (\neg f \mid p(w) \wedge \text{Context})$ indicates that denial is possible given appropriate contextual information.

The crucial test is cancellation behavior: "The emu is a bird but cannot fly" is perfectly acceptable; the default inference is cancelled by the explicit statement. However, the default still influences processing: encountering *bird* in a neutral context primes flying-related concepts more strongly than non-flying concepts, even for speakers who know about penguins and emus (McRae et al., 1997). This demonstrates that the inference is active by default even though it can be overridden.

Weak Entailments: Arising from cultural connotation level ($\alpha \approx x - 3n$)

Weak entailments are systematic but easily overridden inferences arising from moderate conventionalization. These represent the theoretical innovation required for understanding cultural connotation and semantic prosody phenomena.

Examples include:

- *regime* → authoritarian or oppressive governance
- *cause* → negative outcome
- *utterly* → complete negative state
- *commit* → transgression or binding negative action
- *set in* → undesirable gradual onset

Weak entailments exhibit the following characteristics:

1. *Moderate speaker agreement*: A majority of speakers recognize the association, but substantial variation exists.
2. *High context sensitivity*: The inference is highly dependent on discourse context and easily overridden.
3. *Easy cancellation*: Denial does not produce contradiction or even strong markedness.
4. *Distributional consequences*: While easily overridden in individual cases, the inference produces systematic corpus skews.

Formally, weak entailments can be represented as probabilistic inferences:

If $\alpha(f) \approx x - 3n$, then $P(f \mid p(w)) > 0.5$ but $P(f \mid p(w) \wedge \text{Context})$ is highly variable

This captures the insight that weak entailments create systematic expectations (probability greater than chance) that are nevertheless highly sensitive to context and easily overridden.

The crucial distinction between weak entailments and mere associations lies in their systematicity and cross-speaker consistency. Private associations (such as "dog reminds me of my childhood pet") show minimal intersubjective agreement ($\alpha < 0.5$) and no systematic patterns across speakers. Weak entailments, by contrast, show moderate but consistent agreement and generate predictable patterns in corpus data, experimental measures, and computational modeling. This systematicity makes them amenable to theoretical treatment despite their defeasibility.

Consider the distinction between *cause* and *foster*. Both are causative verbs, but *cause* carries a weak entailment toward negative outcomes while *foster* carries a weak entailment toward positive outcomes. These entailments can be overridden ("cause celebration," "foster

resentment"), but they create systematic distributional skews. The skews arise because the weak entailments align with discourse pragmatics: we more frequently discuss forceful causation of harm than forceful causation of benefit, and we more frequently discuss nurturing of positive developments than nurturing of negative ones.

The Mechanism Linking Conventionalization to Entailment Strength

Why do higher levels of conventionalization generate stronger entailments? The answer lies in how conventionalization affects semantic representation and processing. When a particular semantic feature or inference becomes highly conventionalized—when the vast majority of speakers consistently associate it with a word—it becomes integrated into the core representation of that word's meaning.

This integration occurs through repeated exposure to usage patterns across diverse contexts. As speakers encounter consistent associations between a word and certain features, these features become activated more automatically and reliably when the word is processed. At the neural level, this corresponds to strengthening of synaptic connections between the lexical representation and the featural representation (Pulvermüller, 2013).

We can model this formally using a spreading activation framework. Let $A(w, t)$ represent the activation level of word w at time t , and $A(f, t)$ the activation level of feature f . The influence of w on f can be modeled as:

$$\frac{dA(f, t)}{dt} = \alpha(f) \cdot A(w, t) - \delta \cdot A(f, t)$$

where $\alpha(f)$ is the agreement score (reflecting conventionalization level) and δ is a decay parameter. This equation captures the idea that higher conventionalization (α closer to 1) produces stronger and more rapid activation of the feature when the word is processed.

At the highest levels of conventionalization (strong entailments), the integration becomes so complete that the inference is no longer cognitively separable from the word's meaning. Denying the inference requires either rejecting the word's applicability altogether or interpreting the word as having a different meaning. For instance, denying that a bachelor is unmarried forces us to either reject the claim that the person is a bachelor or to interpret *bachelor* in a non-standard sense (perhaps "bachelor's degree" or an honorific title). The feature has become constitutive of the word's core meaning.

At lower levels of conventionalization (default and weak entailments), the feature remains partially separable from core meaning. Speakers recognize the association as typical or expected but do not treat it as definitional. The inference can be overridden when contextual information provides sufficient motivation, but in the absence of such information, the default expectation prevails. The processing system has learned a probabilistic association: encountering the word typically predicts the feature, but the prediction can be updated based on context.

This account predicts a correlation between agreement level and cancellability: the higher the level of intersubjective agreement on an inference, the more difficult it should be to cancel that inference without producing contradiction or marked usage. We can formalize this prediction:

$$\text{Cancellability}(f) = k \cdot (1 - \alpha(f))$$

where k is a scaling constant. This predicts that features with very high agreement (α close to 1) should be essentially non-cancellable (strong entailments), features with high agreement ($\alpha \approx 0.8$) should be cancellable but marked (default entailments), and features with moderate agreement ($\alpha \approx 0.6$) should be easily cancellable (weak entailments).

Preliminary experimental evidence supports this prediction. Features rated as more "central" or "definitional" by speakers are also more difficult to deny without perceived contradiction (Hampton, 1995). Moreover, processing studies show that violations of higher-agreement features produce stronger N400 effects (indicating semantic anomaly detection) than violations of lower-agreement features (Federmeier & Kutas, 1999). The gradient nature of these effects—rather than categorical differences—supports the continuous gradient model over categorical distinctions.

Compositional Behavior of Different Entailment Types

An important question concerns how different types of entailments behave under semantic composition. When two words combine, how do their entailments interact?

For strong entailments, composition follows classical logical rules. If w_1 strongly entails f_1 and w_2 strongly entails f_2 , then the composition $[w_1 w_2]$ entails both f_1 and f_2 (assuming compositionality):

$$p(w_1) \models f_1 \wedge p(w_2) \models f_2 \Rightarrow p([w_1 w_2]) \models f_1 \wedge f_2$$

For example, "female bachelor" would entail both "female" (from *female*) and "unmarried adult male" (from *bachelor*), creating a contradiction that renders the phrase semantically anomalous.

For default entailments, composition is more complex. Default entailments can be overridden not only by explicit contextual information but also by composition with words whose entailments conflict. Consider "flightless bird." The default entailment from *bird* (can fly) is cancelled by composition with *flightless*, which explicitly denies that entailment. The composition is acceptable because the modifier overrides the default:

$$p([w_1 w_2]) \Rightarrow f \text{ unless } f_2 = \neg f$$

This captures the intuition that default entailments project through composition but can be cancelled by explicit denial within the compositional structure.

For weak entailments, composition permits more complex interactions. Weak entailments can reinforce each other when features align, or they can neutralize each other when features conflict. Consider "utterly cause devastation"—both *utterly* and *cause* carry weak entailments toward negativity, and these reinforce each other, making the negative outcome more salient. Conversely, "foster peace" combines a facilitative verb with a positive outcome, reinforcing positive expectations.

More interesting are cases where weak entailments conflict: "cause celebration" or "foster resentment." Here, composition produces acceptable but somewhat marked expressions. The markedness arises from the tension between conflicting weak entailments, but because these entailments are weak rather than strong, the tension does not produce contradiction. Instead, it creates a rhetorical effect—the expression draws attention to an atypical instance of causation or fostering.

We can model weak entailment composition probabilistically:

$$P(f \mid [w_1 \ w_2]) = P(f \mid w_1) \cdot P(f \mid w_2) \cdot \lambda + (1 - \lambda) \cdot P(f \mid \text{Context})$$

where λ represents the weight given to lexical versus contextual information. When weak entailments from w_1 and w_2 align (both predicting f or both predicting $\neg f$), the combined probability is strengthened. When they conflict, the result depends on contextual weighting.

Summary: The Entailment Typology

Table 1 summarizes the relationship between conventionalization level, entailment type, and key diagnostic properties.

Table 1*Conventionalization Levels and Entailment Types*

Level	Agreement Score	Entailment Type	Cancellability	Example
Reference	$x (\rightarrow 1.0)$	Strong	Contradictory	<i>dog</i> $\rightarrow$ animal
Denotation	$x - n$	Strong	Contradictory	<i>bachelor</i> $\rightarrow$ unmarried
Sense	$x - 2n$	Default	Marked but acceptable	<i>bird</i> $\rightarrow$ flies
Cultural Connotation	$x - 3n$	Weak	Easily overridden	<i>regime</i> $\rightarrow$ oppressive
Private Connotation	$x - 4n$	None	N/A	<i>dog</i> $\rightarrow$ childhood memory

This typology provides the theoretical foundation for understanding how meaning components at different conventionalization levels constrain inference and processing. Strong entailments function as necessary inferences, default entailments as typical but overridable expectations, and weak entailments as systematic but easily cancelled tendencies. The gradient of entailment strength corresponds directly to the gradient of conventionalization, providing a unified account of how semantic structure constrains both meaning and inference.

Cultural Versus Private Connotation

The distinction between weak entailments and mere associations corresponds to a crucial but neglected distinction between two types of phenomena that have been conflated under the label "connotation": *cultural connotation* and *private connotation*. Failure to distinguish these has led to systematic undertheorization of connotative meaning in formal semantics. This section

diagnoses the conflation problem and demonstrates the theoretically significant differences between these phenomena.

The Conflation Problem and Its Consequences

Traditional discussions of connotation in linguistics often treat it as a unitary phenomenon comprising subjective, affective, or evaluative associations that vary across speakers and contexts. Lyons (1977, p. 176) characterizes connotative meaning as the "communicative value that an expression has by virtue of what it refers to, over and above its purely conceptual content." This characterization lumps together several distinct phenomena:

1. Systematic tendencies for words to appear in particular semantic or evaluative contexts (e.g., *regime* in authoritarian contexts)
2. Social or register-based associations (e.g., *father* as formal versus *dad* as casual)
3. Subjective emotional responses (e.g., *moist* as unpleasant for some speakers)
4. Individual autobiographical associations (e.g., *beach* evoking childhood vacations)

While all of these phenomena involve associations beyond core denotational meaning, they differ fundamentally in their level of systematicity and intersubjective agreement. The first two types show considerable cross-speaker consistency and generate predictable patterns in corpus data. The latter two show much more individual variation and lack systematic distributional signatures.

This conflation has had serious consequences for semantic theory. Observing that connotative associations vary across speakers and contexts, theorists have concluded that connotation lies outside the proper domain of semantic theory and belongs instead to pragmatics, sociolinguistics, or individual psychology (Cruse, 1986). This conclusion would be warranted for purely private associations but is premature for systematic, community-level patterns.

The problem is exacerbated by the traditional lexical-encyclopedic boundary. If we assume that lexical meaning must be categorical, universal, and stable while encyclopedic knowledge is gradient, variable, and context-dependent, then any meaning component showing variation gets classified as encyclopedic and thus outside semantics proper. But this reasoning is circular: it assumes the categorical boundary it purports to discover.

Cultural Connotation: Systematic and Theorizable

Cultural connotation comprises associations that show moderate to high intersubjective agreement within a speech community ($\alpha = 0.50 - 0.70$). These associations arise through conventional patterns of language use and are learned through exposure to how words are actually employed in discourse. Crucially, cultural connotations generate weak entailments: they lead to systematic expectations that influence processing, production, and interpretation, even if these expectations can be overridden by explicit contextual information.

Consider the example of *regime*. Corpus analysis across multiple varieties of English consistently shows that *regime* appears predominantly in contexts describing authoritarian, oppressive, or otherwise negatively evaluated systems of governance (Hunston, 2007). While *regime* can be used neutrally (as in "exercise regime" or "dietary regime"), political uses are marked by negative evaluation. Native speakers, when asked to judge the naturalness of phrases like "democratic regime" versus "authoritarian regime," show systematic preferences that align with corpus frequencies, even if they cannot consciously articulate the association.

This pattern exhibits several key properties that distinguish cultural connotation from both strong semantic content and private association:

****Cross-speaker consistency**:** The negative prosody is not idiosyncratic but shows up consistently across speakers and corpora. We can measure this consistency through corpus analysis across diverse text types, acceptability judgments from multiple participants, and

priming studies showing consistent facilitation patterns. If we measure the agreement score α for the feature "authoritarian/oppressive governance" associated with *regime*, we find values typically in the range 0.50-0.65—moderate but clearly above chance.

Systematicity: The pattern is not random but reflects structured semantic features (authoritarianism, oppression, system control) rather than arbitrary co-occurrence. The association makes sense given the semantic profile of *regime* (systematic, formalized control) and the pragmatic salience of authoritarian systems in political discourse. This systematicity distinguishes cultural connotation from spurious corpus correlations.

Processing effects: The association influences online processing, as demonstrated by priming studies and reading time measures. When participants read sentences containing *regime*, processing is facilitated for subsequently presented words related to authoritarianism or oppression, even when the *regime* sentence itself was neutral (Stewart, 2010). This demonstrates that the association is activated automatically during comprehension.

Gradedness: The strength of the association varies continuously with context rather than being categorical. In strongly supportive contexts ("democratic regime based on popular sovereignty"), the negative association is weakened or neutralized. In neutral contexts ("the regime"), the association is moderately activated. In reinforcing contexts ("oppressive regime"), the association is strengthened. This gradient behavior aligns with the weak entailment framework.

Cultural variation: While systematic within a speech community, the specific associations may vary across different language communities or historical periods. The negative association of *regime* is stronger in contemporary Western English than in other varieties or historical periods. This variation demonstrates that the association is conventional rather than inherent to the word's core meaning.

The mechanism underlying cultural connotation is straightforward: through repeated exposure to patterns of usage, speakers extract statistical regularities about the contexts in which words typically appear. These regularities become conventionalized to varying degrees, with higher-frequency patterns achieving higher levels of intersubjective agreement. The resulting expectations function as weak entailments, generating default inferences that can be overridden but nevertheless influence language processing and production.

We can model this learning process formally. Let C represent the set of contexts in which a word w appears in a corpus, and let each context $c_i \in C$ have an associated feature value v_i for feature f (e.g., evaluative valence). The conventionalized association strength can be estimated as:

$$\alpha(f) = \frac{1}{|C|} \sum_{c_i \in C} v_i \cdot \text{weight}(c_i)$$

where $\text{weight}(c_i)$ reflects factors like frequency, recency, and salience. As speakers are exposed to consistent patterns (e.g., *regime* predominantly appearing with negative political contexts), the estimated $\alpha(f)$ converges toward the population-level conventionalization. Individual speakers may converge at different rates or to slightly different values depending on their exposure, explaining individual variation around the community mean.

Unlike strong entailments derived from definitional features, cultural connotations reflect distributional learning from actual usage patterns. The association between *regime* and authoritarianism is not part of the word's dictionary definition but emerges from the statistical distribution of its use in discourse. This distinguishes cultural connotation from denotation while explaining why it nevertheless shows systematicity.

Private Connotation: Idiosyncratic and Non-Theorizable

Private connotation, by contrast, comprises associations arising from individual experience and showing minimal intersubjective agreement ($\alpha < 0.50$). These associations are genuinely idiosyncratic, varying dramatically across individuals based on personal history, emotional experiences, and autobiographical memory.

For instance, the word *hospital* might evoke positive associations for one speaker (remembering the birth of a child), negative associations for another (remembering a parent's illness), and neutral or no particular association for a third. These associations may be emotionally powerful for the individuals who hold them, but they show no systematic pattern across speakers and generate no predictable distributional signatures in corpus data.

Private connotations do not generate entailments in any meaningful sense. They do not create expectations about how words will be used or understood in discourse. They do not influence processing in ways that generalize across speakers. And they cannot be learned from distributional patterns in language use because they reflect extra-linguistic experience rather than conventional patterns of usage.

Consider the difference in learning mechanisms. Cultural connotations can be acquired purely from linguistic input—observing that *regime* tends to appear in negative political contexts, even without direct experience of authoritarian governments. A language learner could extract this pattern from reading news articles or textbooks. Private connotations, however, require personal experience. A computational model trained on corpus data would not acquire the association between *hospital* and any particular individual's autobiographical memories, because these associations are not reflected in the distributional patterns of the corpus.

Formally, if we attempt to estimate $\alpha(f)$ for a private association f across the population, we get:

$$\alpha(f_{\text{private}}) = \frac{1}{|\mathcal{S}|} \sum_{s \in \mathcal{S}} \mathbb{1}[\text{speaker } s \text{ has association } f] \approx 0$$

where $\mathbb{1}[\cdot]$ is an indicator function. Because each individual's private associations are unique, summing across the population yields values approaching zero. This low agreement score explains why private connotations cannot generate systematic corpus patterns or support theoretical analysis at the community level.

Importantly, the distinction between cultural and private connotation is not categorical but gradient. Some associations show intermediate levels of agreement—perhaps shared within subgroups but not across the entire speech community. For instance, certain professional jargon or regional dialects may show moderate agreement within specific communities but low agreement in the broader population.

The key diagnostic is systematicity: do the associations generate predictable patterns in language use, or are they purely idiosyncratic? Cultural connotations, even when specific to subgroups, still show systematic patterns within those groups and can be learned from exposure to the group's linguistic practices. Private connotations show no such systematicity and cannot be learned from distributional evidence alone.

Why the Distinction Matters for Semantic Theory

The failure to distinguish cultural from private connotation has led to a systematic undertheorization of connotative meaning in formal semantics. Observing that connotative associations vary across speakers and lack the categorical character of strong entailments,

theorists have concluded that all connotation lies outside semantic theory's proper scope. This conclusion rests on the implicit assumption that all connotation is private.

Once we recognize that cultural connotation exhibits sufficient systematicity to warrant theoretical treatment, several new research questions emerge:

Question 1: Relationship to other non-truth-conditional meaning. How do cultural connotations relate to presuppositions, conventional implicatures, and expressives? All three phenomena contribute to meaning without affecting truth conditions, but they differ in their conventionalization levels and cancellability properties. Presuppositions typically show very high conventionalization (near denotation level), conventional implicatures show high to moderate conventionalization (sense to cultural connotation level), and cultural connotations show moderate conventionalization (weak entailment level). Understanding these relationships could lead to a unified theory of non-truth-conditional meaning organized along the conventionalization gradient.

Question 2: Compositional semantics of weak entailments. Do weak entailments compose systematically, and if so, according to what principles? When *utterly* (with a weak entailment toward negativity) combines with *cause* (also with a weak entailment toward negativity), do the weak entailments reinforce each other? What formal mechanisms govern this interaction? These questions parallel longstanding debates about presupposition projection but have not been systematically explored for connotative meaning.

Question 3: Acquisition mechanisms. Do children acquire cultural connotations through the same learning mechanisms as core denotational features, or do different types of input and processing support different types of meaning? The distributional learning hypothesis suggests that cultural connotations should be acquirable from statistical patterns in linguistic input, but empirical investigation is needed to confirm this and to identify developmental trajectories.

Question 4: Diachronic semantic change. How do cultural connotations change over time? The amelioration of *nice* from "foolish" to "pleasant" and the pejoration of *regime* from "system of government" to "authoritarian system" likely involved shifts in conventionalization levels. Tracking these changes systematically could reveal general principles of semantic evolution and the mechanisms by which weak entailments become stronger or weaker over time.

Question 5: Formalization and representation. How should we formally represent weak entailments and gradations in entailmental strength? This requires extending current semantic formalisms to handle probabilistic or graded inferences while maintaining compositionality and explanatory power. Several approaches are possible: probabilistic semantics (Goodman & Lassiter, 2015), fuzzy logic frameworks (Zadeh, 1965), or enriched type theories with gradient features.

Progress on these questions would significantly expand the empirical coverage of semantic theory, enabling engagement with a wider range of systematic linguistic phenomena while maintaining theoretical rigor. The gradient framework provides the conceptual foundation for such expansion by demonstrating that systematicity comes in degrees and that moderate systematicity (cultural connotation) warrants theoretical attention even when it does not achieve the universality of strong entailments.

Semantic Prosody Reconsidered

The framework developed above provides new tools for understanding semantic prosody, a phenomenon that has resisted satisfactory theoretical treatment despite extensive corpus documentation. Semantic prosody refers to the tendency for certain words to appear consistently in contexts with particular semantic or evaluative characteristics, even when these characteristics are not part of the word's core denotational meaning (Louw, 1993; Sinclair, 1991; Stubbs, 2001).

This section demonstrates how reconceptualizing prosody as cultural connotation generating weak entailments resolves several longstanding theoretical puzzles.

The Semantic Prosody Puzzle

Consider the verb *cause*. Corpus analysis across diverse text types consistently reveals that *cause* appears predominantly with complement clauses describing negative or undesirable outcomes. Stubbs (2001) documents this pattern extensively: *cause* combines readily with *problem, damage, concern, cancer, death, and trouble* (negative outcomes), but rarely appears with *benefit, improvement, happiness, or success* (positive outcomes). Quantitative analysis of the Bank of English corpus shows that approximately 70% of transitive uses of *cause* involve negatively valenced direct objects.

Similar patterns appear with numerous other words. The verb *commit* shows strong prosodic affinity with crime, error, or transgression contexts (*commit murder, commit suicide, commit adultery, commit perjury*) but appears marked with neutral or positive objects. The intensifier *utterly* patterns with negative adjectives (*utterly destroyed, utterly devastated, utterly hopeless*) but sounds odd with positive ones. The phrasal verb *set in* collocates overwhelmingly with undesirable states (*rot set in, decay set in, despair set in*).

These patterns present several puzzles for traditional semantic theory:

Puzzle 1: Systematicity without consciousness. The patterns are highly systematic in corpus data yet largely invisible to native speakers' conscious intuitions. Speakers cannot usually articulate these prosodic preferences when asked directly but demonstrate them consistently in production and comprehension. This dissociation between behavioral systematicity and metalinguistic awareness requires explanation.

Puzzle 2: Statistical but non-arbitrary. The patterns reflect genuine statistical tendencies in language use, not arbitrary co-occurrence, yet they do not follow from core

denotational meaning in any obvious way. The negativity of *cause*'s typical objects is not entailed by the meaning "bring about" or "produce." The pattern is systematic but appears to lack semantic grounding.

Puzzle 3: Resistance to classical tests. The prosodic associations do not pattern like classical entailments. Saying "The policy caused economic growth" is not contradictory or even particularly marked, yet corpus frequencies show consistent negativity bias. Classical semantic diagnostics (entailment, contradiction, anomaly) fail to capture prosodic patterns, suggesting they constitute a different type of meaning.

Puzzle 4: Cross-linguistic variation. While many prosodic patterns show cross-linguistic parallels, others vary across languages, suggesting conventional rather than natural grounding. The negative prosody of English *cause* is not necessarily replicated in translation equivalents across all languages, indicating that the pattern is not purely a matter of conceptual structure.

Traditional approaches have offered two main accounts, neither fully satisfactory:

Account 1: Absorption model (corpus linguistic approach). Corpus linguists often treat prosody as reflecting pragmatic or cultural knowledge that gets "absorbed" into words through repeated co-occurrence (Hoey, 2005; Sinclair, 1991). Under this view, *cause* has acquired negative associations because it frequently appears with negative outcomes, and through repeated exposure speakers come to associate the word itself with negativity.

This account struggles to explain the patterns' systematicity and psychological reality. If prosody is simply absorbed contextual association, why does it show structured semantic patterns rather than arbitrary co-occurrence? Why does *cause* associate specifically with forcefully imposed negative outcomes rather than with any random subset of negative contexts?

The absorption model describes the correlation but fails to explain why particular words absorb particular types of associations.

Account 2: Pragmatic dismissal (formal semantic approach). Formal semanticists typically dismiss prosody as mere statistical association without theoretical status (Whitsitt, 2005). Under this view, corpus patterns reflect discourse preferences and pragmatic tendencies but do not constitute part of semantic meaning. The negativity of *cause*'s typical objects tells us about what speakers choose to talk about, not about what *cause* means.

This move fails to engage with the phenomenon's systematicity and cross-speaker consistency. The patterns are too robust and too predictable to dismiss as pragmatic noise. Moreover, the patterns influence actual language processing in measurable ways, suggesting they have psychological reality beyond mere statistical description.

Prosody as Cultural Connotation Generating Weak Entailments

The gradient framework suggests a different analysis: semantic prosody reflects cultural connotation operating at moderate conventionalization levels ($\alpha \approx x - 3n$) and generating weak entailments. Under this view, prosodic patterns arise not from absorption of contextual associations but from systematic encoding of semantic features that constrain distributional possibilities through their interaction with contextual semantics and discourse pragmatics.

Consider *cause* more carefully. The verb does not carry an inherent negative prosody. Rather, it encodes a semantic profile involving direct, forceful agentivity—features that naturally align with imposing unwelcome states but create semantic tension with states that subjects typically welcome. When we say "The storm caused damage," the semantics of *cause* (forceful imposition) aligns naturally with the semantics of *damage* (unwelcome state). When we say "The policy caused prosperity," the same features of *cause* create tension with the welcoming nature of *prosperity*.

This tension does not produce contradiction because the association is a weak entailment rather than a strong one. The usage is acceptable because the weak entailment can be overridden by contextual information. However, the tension is reflected in lower corpus frequencies and potential for rhetorical effects. Speakers can use "cause prosperity" when they want to emphasize the forceful, direct nature of the causal relationship, but they more naturally reach for alternatives like "foster prosperity" or "promote prosperity" when describing gradual cultivation of positive outcomes.

Formally, we can represent this as:

$$P(\text{negative} \mid \text{cause}) = \alpha(\text{agentive-forceful}) \cdot P(\text{negative} \mid \text{agentive-forceful}) + (1 - \alpha(\text{agentive-forceful})) \cdot P(\text{negative} \mid \text{discourse})$$

The first term captures the weak entailment: the agentive-forceful profile (with moderate conventionalization $\alpha \approx 0.6$) creates a probabilistic preference for contexts where forceful imposition is conceptually appropriate, which correlates with negative outcomes. The second term captures general discourse preferences for discussing negative causation. The overall distribution is the weighted combination.

This analysis receives support from several sources:

Evidence 1: Entailment patterns. Prosodic features behave like weak entailments rather than strong entailments or pure pragmatic implicatures. They create default expectations that influence processing but can be overridden by explicit contextual information. This aligns with the moderate conventionalization level ($\alpha = 0.50 - 0.70$) proposed for cultural connotation.

Evidence 2: Cross-speaker consistency. Experimental studies find moderate agreement across speakers on prosodic judgments, higher than would be expected for pure statistical association but lower than for definitional features. When asked to rate the naturalness of "*cause*

+ positive outcome" versus "*cause* + negative outcome," speakers show systematic preferences with agreement scores typically around 0.55-0.65, precisely the range predicted for cultural connotation (Blumenthal-Dramé, 2016).

Evidence 3: Systematicity of patterns. Prosodic patterns reflect structured semantic features (agentivity, volitionality, evaluation) rather than arbitrary co-occurrence. The features interact with contextual semantics in predictable ways. *Cause* prefers direct, forceful impositions regardless of whether they are negative, but the intersection with discourse pragmatics (we more often discuss forceful imposition of harm) produces the observed distributional skew.

Evidence 4: Compositional interaction. Prosodic features interact with compositional semantics, influencing interpretation at the sentence level. This suggests integration with semantic representation rather than purely pragmatic status. The prosodic features of *cause* interact with the evaluative properties of complement predicates to generate expectations about sentence meaning that are probabilistic but systematic.

Evidence 5: Language model extraction. Neural language models successfully extract prosodic patterns from distributional data (Devlin et al., 2019), suggesting they reflect systematic features rather than noise. Models trained purely on corpus distributions develop representations that capture prosodic associations, demonstrating that these patterns are learnable from usage statistics. Moreover, models show the same pattern of flexible deployment—readily producing both typical uses ("cause damage") and counterexamples ("cause celebration")—that human speakers show.

Semantic Profiles and Distributional Patterns

The key theoretical move is recognizing that words encode *semantic profiles*: structured constellations of features that do not reduce to simple truth conditions but nevertheless systematically constrain usage possibilities. For *cause*, the profile includes agentivity (acting

upon), directness (immediate connection), and forcefulness (overcoming resistance). These features make *cause* naturally compatible with contexts describing imposition of unwelcome states while creating semantic tension with contexts describing positive outcomes.

We can formalize semantic profiles as feature bundles with associated conventionalization levels:

$$\text{Profile}(w) = \{(f_1, \alpha_1), (f_2, \alpha_2), \dots, (f_n, \alpha_n)\}$$

For **cause**:

$$\text{Profile}(\text{cause}) = \{(\text{agentive}, 0.85), (\text{direct}, 0.80), (\text{forceful}, 0.75), (\text{negative-outcome}, 0.60)\}$$

The first three features show high conventionalization (sense level), constituting relatively stable semantic content. The fourth feature shows moderate conventionalization (cultural connotation level), constituting a weak entailment. The profile as a whole constrains usage by making certain contexts more or less compatible, but because the evaluative feature has only moderate conventionalization, counterexamples remain fully acceptable.

Crucially, semantic profiles occupy intermediate conventionalization levels ($\alpha = 0.50 - 0.70$ for the most variable features). They show sufficient intersubjective agreement to generate systematic corpus patterns and processing effects but insufficient agreement to achieve the status of classical entailments. This explains their "intuition-resistant" character: speakers demonstrate knowledge of prosodic patterns through consistent production and comprehension behavior while struggling to consciously articulate them.

The patterns exist at a level of conventionalization that influences processing automatically without requiring conscious awareness. This dissociation between behavioral effects and metalinguistic access is predicted by dual-process theories of cognition (Evans,

2008). Features with very high conventionalization (strong entailments) are accessible to both implicit processing and explicit reflection. Features with moderate conventionalization (weak entailments) influence implicit processing but remain below the threshold for reliable explicit access.

Case Study: *Cause* versus *Foster*

A detailed comparison of *cause* and *foster* illustrates how semantic profiles generate contrasting prosodic patterns. Both are causative verbs, but they encode different profiles and consequently show different distributional patterns.

For **cause**:

Profile(*cause*) = {(agentive, 0.85), (direct, 0.80), (forceful, 0.75), (immediate, 0.70)}

For **foster**:

Profile(*foster*) = {(facilitative, 0.85), (gradual, 0.80), (nurturing, 0.75), (indirect, 0.70)}

These contrasting profiles predict the observed distributional patterns. *Cause* appears predominantly with negative outcomes because negative events (harm, damage, death) are prototypically conceptualized as forcefully imposed by agents. *Foster* appears predominantly with positive outcomes because positive developments (growth, understanding, cooperation) are prototypically conceptualized as gradually nurtured through supportive conditions.

Critically, both verbs can override their typical patterns when the profile features remain satisfied. "Cause celebration" is acceptable when the context supports forceful, direct, immediate bringing-about of a celebratory state. "Foster resentment" is acceptable when the context supports gradual, indirect nurturing of resentful feelings. The profiles constrain but do not absolutely determine usage.

Corpus evidence confirms this analysis. COCA (Davies, 2008-) yields numerous counterexamples for both verbs:

For *cause*: "cause excitement," "cause celebration," "cause joy," "cause improvement"

For *foster*: "foster dependence," "foster resentment," "foster inequality," "foster complacency"

These counterexamples are not marginal or ironically marked. They straightforwardly describe situations where the profile features apply to atypical outcomes. The fact that such uses are comprehensible without contradiction definitively demonstrates that the evaluative associations are weak entailments rather than strong semantic content.

Resolving the Four Puzzles

The profile-based analysis resolves the four puzzles identified earlier:

Resolution of Puzzle 1 (Systematicity without consciousness). The dissociation between behavioral systematicity and metalinguistic awareness follows from moderate conventionalization levels. Weak entailments (cultural connotations) operate at agreement levels ($\alpha \approx 0.60$) sufficient to influence automatic processing but insufficient for reliable conscious access. Speakers demonstrate the patterns behaviorally because their processing systems have learned the statistical regularities, but they struggle to articulate them because the features have not achieved the high conventionalization necessary for metalinguistic awareness.

Resolution of Puzzle 2 (Statistical but non-arbitrary). The patterns are statistically robust because they reflect genuine semantic structure (profiles), but they appear non-arbitrary because the profiles interact with discourse pragmatics and conceptual structure. The negative tendency of *cause* is not arbitrary—it follows from the agentive-forceful profile interacting with the fact that negative events are more often conceptualized as requiring forceful agents. The pattern is both statistically measurable and semantically grounded.

Resolution of Puzzle 3 (Resistance to classical tests). Prosodic associations resist classical semantic tests because they are weak entailments rather than strong ones. Classical tests (contradiction, anomaly) diagnose strong entailments, but weak entailments behave differently—they influence probability distributions without creating categorical constraints. The appropriate test is not contradiction but distributional skew and processing facilitation.

Resolution of Puzzle 4 (Cross-linguistic variation). Cross-linguistic variation in prosodic patterns is predicted by the framework. While semantic profiles may show some cross-linguistic stability (direct causation is conceptually similar across languages), the exact conventionalization level of particular features can vary. What is moderately conventionalized in one language ($\alpha \approx 0.60$) may be more or less conventionalized in another ($\alpha \approx 0.70$ or 0.50), producing different strength prosodic patterns.

Theoretical Implications

Reanalyzing semantic prosody as cultural connotation generating weak entailments has several important implications for semantic theory:

Implication 1: Validation of prosody as semantic phenomenon. Prosodic patterns reflect systematic aspects of lexical meaning that interact with compositional semantics and influence interpretation. They are not merely statistical artifacts or pragmatic effects but genuine components of word meaning that happen to occupy moderate positions on the conventionalization gradient. This validates prosody as a legitimate object of semantic inquiry.

Implication 2: Explanation of paradoxical properties. The apparently paradoxical properties of prosody—systematic yet intuition-resistant, statistical yet structured, meaningful yet non-truth-conditional—all follow naturally from intermediate conventionalization levels. The systematicity reflects genuine conventionalization above random chance. The intuition-resistance reflects insufficient conventionalization for conscious access. The statistical nature reflects the

gradient and probabilistic character of weak entailments. The structured patterns reflect the semantic content of underlying profiles.

Implication 3: Connection to other semantic phenomena. Prosody research connects productively to other domains of semantic theory including presupposition, conventional implicature, and expressive meaning. All of these phenomena may reflect different types of conventionalized meaning operating at different gradient levels. Presuppositions might occupy higher conventionalization levels than prosody (sense level rather than cultural connotation level), while expressives might occupy similar or lower levels. Mapping these relationships systematically could lead to a unified theory of non-truth-conditional meaning organized along the conventionalization gradient.

Implication 4: Methodological opportunities. If prosodic features reflect moderate conventionalization, they should be measurable through agreement studies, manipulable in experimental contexts, and learnable from distributional evidence. These predictions enable empirical testing of the framework using converging evidence from corpus analysis, psycholinguistic experimentation, and computational modeling. The framework thus transforms prosody from a curious descriptive phenomenon into a theoretically significant testing ground for claims about meaning representation.

Epistemological Applications: The Gradient and Argument Dynamics

Beyond resolving puzzles in lexical semantics, the gradient framework provides theoretical tools for understanding a seemingly unrelated phenomenon: the epistemology of disagreement. The same gradient of conventionalization that distinguishes types of lexical meaning also predicts patterns in argumentative discourse. This section develops this connection, demonstrating how semantic structure constrains the dynamics of belief revision and consensus formation.

The Core Prediction

The fundamental insight is that the level of conventionalization (intersubjective agreement) on the meanings of terms employed in an argument predicts whether that argument will converge toward consensus or diverge toward polarization. We can formalize this as:

Convergence Principle: For any argumentative exchange involving term w with feature f at conventionalization level $\alpha(f)$:

- If $\alpha(f) \geq x - n$ (reference/denotation level): Arguments converge toward consensus through evidence and reasoning.
- If $\alpha(f) \approx x - 2n$ (sense level): Arguments become unproductive; interlocutors talk past each other without convincing.
- If $\alpha(f) \approx x - 3n$ (cultural connotation level): Arguments actively widen differences and increase polarization.
- If $\alpha(f) < x - 3n$ (private connotation level): Arguments are impossible because terms lack shared meaning.

This principle generates precise predictions about when arguments will be productive versus counterproductive, and why certain domains of inquiry converge on consensus while others remain perpetually contested.

Arguments at High Conventionalization Levels

When arguments employ terms with very high intersubjective agreement ($\alpha \geq x - n$), the shared semantic foundation enables productive dispute resolution. Consider an argument about whether a particular animal is a dog. Both parties agree on what *dog* means (reference level: four-legged canine mammal). The disagreement concerns not the meaning but the facts: Is this creature a canine? Does it have four legs? These are empirical questions that can be resolved through observation and evidence.

More subtly, consider arguments involving denotation-level features. An argument about whether John is a bachelor can be resolved by establishing the facts: Is John unmarried? Is John an adult? Is John male? Because **bachelor** $\rightarrow$ unmarried is a strong entailment with very high agreement ($\alpha \geq 0.90$), both parties share the inference and can focus on determining whether the necessary conditions are satisfied.

The convergence mechanism operates through shared inference rules. When both parties agree that $p \models q$ (strong entailment), they can jointly reason: If we establish p , then we must accept q . Disagreement about conclusions can be traced back to disagreement about premises, which can then be investigated empirically. The shared semantic foundation provides common ground for rational persuasion.

Formally, let A and B be two arguers with belief sets $\mathcal{B}_A$ and $\mathcal{B}_B$. If they both accept $\alpha(f) \geq x - n$ for relevant feature f , then:

$$p \in \mathcal{B}_A \wedge p \models_f q \Rightarrow (A \text{ can rationally persuade } B \text{ that } q \text{ by establishing } p)$$

The persuasion succeeds because B shares A 's acceptance of the inference $p \models_f q$. The high conventionalization ensures that the inference is not contested.

This explains why scientific arguments about empirical matters, when they employ well-defined technical terms, often converge. Terms like **electron**, **DNA**, **photosynthesis** have highly conventionalized meanings within scientific communities ($\alpha \geq 0.85$). Arguments about whether a particular particle is an electron can be resolved by establishing whether it satisfies the defining criteria (mass, charge, spin). The shared understanding of what **electron** means provides the foundation for productive disagreement about particular cases.

Arguments at Intermediate Conventionalization Levels

When arguments employ terms at the sense level ($\alpha \approx x - 2n$), with agreement scores around 0.70-0.85, a different dynamic emerges. At this level, speakers agree on core denotation but may organize semantic fields differently, prioritize different features, or have different intuitions about borderline cases.

Consider an argument about whether a tomato is a fruit or a vegetable. Botanically, *fruit* is defined by reproductive function (mature ovary containing seeds), placing tomatoes clearly in the fruit category. Culinarily, *fruit* is organized by taste and culinary use (sweet, eaten raw or in desserts), placing tomatoes in the vegetable category. Both parties may understand both definitions, but they prioritize different semantic organizations—biological versus culinary.

This creates a characteristic pattern: the argument becomes unproductive because interlocutors talk past each other. The botanist argues, "Tomatoes have seeds and develop from ovaries, therefore they are fruits." The chef responds, "Tomatoes are savory and used in salads and main courses, therefore they are vegetables." Both arguments are valid given their respective semantic organizations, but neither persuades the other because they are employing *fruit* at different positions in the semantic field.

Formally, if $\alpha(f_1) \approx x - 2n$ and $\alpha(f_2) \approx x - 2n$ where f_1 and f_2 are competing organizational features:

$A \text{ organizes } w \text{ by } f_1 \wedge B \text{ organizes } w \text{ by } f_2 \Rightarrow \text{argument becomes unproductive}$

Neither party can persuade the other by providing evidence, because they do not disagree about evidence—they disagree about which semantic organization is appropriate. This is a metalinguistic negotiation about meaning rather than an empirical dispute about facts.

This pattern is pervasive in philosophical argumentation. Consider disputes about whether acts of civil disobedience are "legal" or "illegal." Legal positivists organize *legal* around formal institutional criteria (conformity with legislated statutes), making civil disobedience clearly illegal. Natural law theorists organize *legal* around moral legitimacy and just authority, potentially making unjust laws "illegal" in a deeper sense. Both parties understand both criteria, but they prioritize different semantic features in organizing the concept.

The unproductiveness arises because sense-level features, while showing high agreement, still permit enough variation that different speakers can organize semantic space differently. Agreement scores around 0.75 mean that 25% of speakers may have different intuitions, and even among the 75% who agree, there may be differences in feature weighting that become apparent in borderline cases.

Arguments at Low Conventionalization Levels

When arguments employ terms at the cultural connotation level ($\alpha \approx x - 3n$), with agreement scores around 0.50-0.70, the dynamics change dramatically. Rather than becoming merely unproductive, arguments actively polarize—they widen rather than narrow differences between interlocutors.

Consider political arguments about "freedom." The term *freedom* has high denotation-level agreement (roughly: absence of constraint), but substantial variation at the sense and cultural connotation levels. Different political traditions emphasize different aspects: freedom from government interference (classical liberalism), freedom from economic necessity (socialism), freedom to participate in collective self-governance (republicanism), freedom from social domination (feminism).

When political opponents argue about whether a policy enhances or restricts "freedom," they are often employing the term with different cultural connotations ($\alpha_A(f_1) \neq \alpha_B(f_2)$). The

libertarian argues, "Taxation restricts freedom by taking property through coercion." The socialist responds, "Taxation enhances freedom by liberating people from economic necessity." Both claims are coherent given their respective conceptions of freedom, but neither persuades the other.

Worse, the argument typically polarizes. Each party, failing to persuade the other, concludes that the opponent either does not truly value freedom or is arguing in bad faith. This attribution of ill will or incompetence further entrenches positions. The argument widens rather than narrows the disagreement:

$$\Delta(\mathcal{B}_A, \mathcal{B}_B)_{\text{post-argument}} > \Delta(\mathcal{B}_A, \mathcal{B}_B)_{\text{pre-argument}}$$

where $\Delta(\mathcal{B}_A, \mathcal{B}_B)$ measures the distance between belief sets.

The polarization mechanism operates through several pathways:

Pathway 1: Confirmation bias amplification. When cultural connotations differ, each party selectively attends to evidence that confirms their conception while dismissing evidence supporting the alternative. The libertarian focuses on cases where taxation enabled government overreach; the socialist focuses on cases where taxation enabled social mobility. Each accumulates evidence for their view while remaining unpersuaded by the other's evidence.

Pathway 2: Motivated reasoning about interlocutor. Failure to persuade despite (what seems like) clear arguments leads to negative attributions about the interlocutor. "They must be irrational, biased, or arguing in bad faith." These attributions further reduce openness to persuasion.

Pathway 3: Identity reinforcement. As the argument proceeds, each party's position becomes more strongly tied to their social and political identity. Conceding would require not

just changing a belief but potentially threatening group membership. This identity entrenchment increases resistance to persuasion.

Pathway 4: Semantic divergence. Through the argument, each party's conception of the contested term becomes more sharply defined in opposition to the other's. The cultural connotations diverge further: $|\alpha_A(f) - \alpha_B(f)|$ increases. This makes future communication even more difficult.

The mathematical structure can be modeled as a dynamical system. Let $\alpha_A(f, t)$ and $\alpha_B(f, t)$ represent the conventionalization levels for feature f for arguers A and B at time t . Under argumentative engagement with moderate initial disagreement:

$$\frac{d}{dt} |\alpha_A(f, t) - \alpha_B(f, t)| > 0$$

The gap widens over time rather than narrowing, representing polarization.

This explains the pervasive phenomenon of political polarization in contemporary discourse. Political arguments increasingly employ terms with contested cultural connotations (*freedom, justice, rights, equality, oppression*) where agreement levels fall in the 0.50-0.70 range. Arguments over such terms predictably polarize because the semantic foundation is insufficiently shared to enable convergence.

Scientific Versus Philosophical Discourse

The gradient framework provides a natural explanation for a puzzling asymmetry: why scientific communities tend to converge on theoretical consensus while philosophical communities do not, despite both employing sophisticated reasoning by intelligent practitioners.

The standard explanations appeal to methodological differences: science has empirical testing while philosophy does not, or science is cumulative while philosophy is not. However,

these explanations face counterexamples. Many philosophical disputes concern empirical matters (philosophy of mind, moral psychology), yet convergence remains elusive. Conversely, some scientific disputes persist despite extensive empirical evidence (interpretations of quantum mechanics, theories of consciousness).

The gradient framework suggests a different explanation: the two disciplines operate primarily at different conventionalization levels, which creates different consensus dynamics regardless of methodology.

Scientific discourse: Operating at denotation level

Mature sciences develop technical vocabularies with very high conventionalization levels ($\alpha \geq 0.85$). Terms like **gene**, **atom**, **force**, **entropy** have precise definitions agreed upon within the scientific community. This high agreement enables several crucial features:

1. **Shared inference rules:** When $p \models q$ is a strong entailment with $\alpha \geq 0.85$, all practitioners accept the inference. This enables cumulative reasoning where established results serve as foundations for further work.
2. **Resolvable empirical disputes:** Disagreements about particular cases can be resolved by establishing whether definitional criteria are satisfied. Does this molecule count as a protein? Check whether it is a polymer of amino acids.
3. **Anomaly detection:** When observations conflict with theoretical predictions, the community shares sufficient semantic foundation to recognize genuine anomalies rather than attributing the conflict to different conceptions of terms.
4. **Paradigm formation:** High-conventionalization vocabularies enable what Kuhn (1962) called "normal science"—puzzle-solving within a shared framework. The shared semantic foundation makes it clear what would count as solving a puzzle.

Formally, for scientific community $\mathcal{S}$ with technical term w and feature f :

$$\alpha_S(f) \geq x - n \Rightarrow \text{convergence dynamic: } \frac{d}{dt} \text{Var}(\mathcal{B}_S) < 0$$

where $\text{Var}(\mathcal{B}_S)$ measures variance in beliefs across the community. Over time, belief variance decreases as evidence accumulates and arguments converge.

Philosophical discourse: Operating at sense level

Philosophy, by contrast, often operates at the sense level ($\alpha \approx x - 2n$), with agreement scores around 0.70-0.85. Philosophical terms like *knowledge*, *justice*, *consciousness*, *causation* have core denotation that most agree on, but substantial variation in how semantic fields are organized and which features are prioritized.

Consider *knowledge*. Most philosophers agree on rough necessary conditions (true belief, justification), giving denotation-level agreement. But they disagree about:

- Whether justification requires truth-tracking (reliabilism) or satisfying evidential norms (evidentialism)
- Whether knowledge is analyzable into components or is conceptually primitive
- Whether knowledge requires certainty or merely reliability
- How knowledge relates to action, assertion, and practical reasoning

These are sense-level disagreements about how *knowledge* relates to other concepts in the epistemic domain. Agreement scores fall around 0.70-0.75—high enough for communication but low enough to sustain persistent disagreement.

This generates the characteristic pattern of philosophical disputes:

1. **Talking past each other:** Different theorists organize the semantic field differently, leading to arguments where both sides seem to make valid points without persuading each other.

2. **Persistent pluralism:** Multiple competing theories persist indefinitely because each can claim to capture important features of the concept, just different features.
3. **Borderline case disputes:** Philosophical arguments often focus on borderline or unusual cases (Gettier cases, trolley problems, brain-in-vat scenarios) where sense-level variation becomes most apparent.
4. **Methodological disputes:** Disagreements about proper methodology often reduce to disagreements about which semantic features are most important—conceptual analysis versus empirical investigation, intuition versus theoretical utility.

Formally, for philosophical community $\mathcal{P}$ with contested term w :

$$\alpha_{\mathcal{P}}(f) \approx x - 2n \Rightarrow \text{no convergence: } \frac{d}{dt} \text{Var}(\mathcal{B}_{\mathcal{P}}) \approx 0$$

Belief variance neither increases nor decreases systematically. The community maintains stable pluralism.

This explains several puzzles about philosophical methodology:

Puzzle 1: Why thought experiments polarize rather than resolve. Thought experiments in philosophy often employ terms at sense level, asking what we should say about borderline cases. But because sense-level features show variation ($\alpha \approx 0.75$ means 25% variation), thought experiments typically reveal pre-existing semantic differences rather than resolving them. The Gettier cases show that people organize *knowledge* differently, but they do not converge everyone toward one organization.

Puzzle 2: Why conceptual analysis remains contested. Conceptual analysis attempts to specify necessary and sufficient conditions for concept application. But if concepts are organized along gradients with sense-level features showing moderate variation, there may be no unique

correct analysis. Different analyses capture different ways of organizing semantic space, all with moderate agreement.

Puzzle 3: Why philosophical progress is non-cumulative. Unlike science, where later theories build on earlier established results, philosophy continually revisits fundamental questions. This reflects the instability of sense-level conventionalization. Without high-agreement foundations, each generation must renegotiate basic semantic organizing principles.

Metalinguistic Negotiation and Its Limits

Given these dynamics, when can metalinguistic negotiation—explicit discussion about what terms should mean—successfully resolve disputes?

The gradient framework predicts success conditions:

Success Condition 1: Negotiation succeeds when parties disagree about which features to prioritize among shared features. Both parties recognize features f_1 and f_2 as relevant to term w , but A weights f_1 more heavily while B weights f_2 more heavily. Negotiation can converge on a compromise weighting or clarify that different weightings serve different purposes.

Success Condition 2: Negotiation succeeds when parties can appeal to shared higher-level principles to adjudicate between competing conceptions. Even with sense-level disagreement about *justice*, parties might agree on meta-principles (conceptual utility, explanatory power, reflective equilibrium) that favor one conception over another.

Success Condition 3: Negotiation succeeds when institutionalization can increase conventionalization. A community can stipulate a technical definition that, through repeated use and enforcement, achieves higher agreement. Legal terms often work this way: *contract*, *property*, *tort* have precise legal definitions that became highly conventionalized within legal communities through institutional reinforcement.

Conversely, metalinguistic negotiation fails when:

Failure Condition 1: Parties have genuinely different feature sets due to different background theories or frameworks. A organizes w by features f_1, f_2, f_3 while B organizes it by features f_4, f_5, f_6 , with minimal overlap. No common semantic ground exists for negotiation.

Failure Condition 2: The contested features have very low conventionalization ($\alpha < 0.50$), approaching private connotation. Negotiation cannot create shared meaning where agreement is fundamentally lacking.

Failure Condition 3: Identity or ideological commitments are strongly tied to particular conceptions. Even if parties could agree on semantic grounds which conception is superior, they resist agreement because conceding would threaten group identity or ideological coherence.

This explains patterns in political metalinguistic negotiation. Attempts to negotiate shared meanings for *freedom, justice, rights* often fail because:

1. Different political traditions organize these concepts around genuinely different feature sets (Failure Condition 1)
2. Cultural connotations show only moderate agreement, insufficient for stable shared meaning (Failure Condition 2)
3. Conceptions are strongly tied to political identities, creating resistance to compromise (Failure Condition 3)

Case Study: The Abortion Debate

The abortion debate illustrates how conventionalization levels shape argument dynamics and explain persistent polarization. The central contested term is *person* (or *human being*), and disagreement concerns when personhood begins.

At the denotation level, both sides agree on core features: persons are entities with moral status, deserving of rights and protections. Agreement on this core is very high ($\alpha \geq 0.90$). The

disagreement occurs at the sense level: which features are necessary and sufficient for personhood?

Pro-life advocates often emphasize biological continuity: a human organism with complete human DNA, developing along a continuous trajectory from conception. This gives $\alpha_{\text{pro-life}}(\text{genetic-humanity}) \approx 0.85$ within that community.

Pro-choice advocates often emphasize psychological criteria: consciousness, sentience, capacity for self-awareness, viability. This gives $\alpha_{\text{pro-choice}}(\text{psychological-capacity}) \approx 0.85$ within that community.

These are sense-level features with high agreement **within** communities but substantial disagreement **between** communities. Overall population agreement falls to $\alpha_{\text{population}}(f) \approx 0.50 - 0.60$, placing the contested features at cultural connotation level.

This generates the predicted polarization dynamic:

1. **Arguments talk past each other:** Pro-life advocates point to genetic continuity from conception; pro-choice advocates point to absence of consciousness in early embryos. Both arguments are valid given respective semantic organizations, but neither persuades the other.
2. **Evidence accumulation without convergence:** Advances in embryology, neuroscience, and neonatology provide more detailed empirical information, but this information is interpreted through different semantic frameworks and fails to resolve the dispute.
3. **Polarization over time:** Rather than moderating positions, sustained argument often entrenches them further. Each side develops increasingly sophisticated arguments for their conception, making positions more theoretically elaborated and identity-bound.
4. **Metalinguistic negotiation fails:** Attempts to negotiate what *person* should mean fail because:

- The feature sets are genuinely different (genetic-biological vs. psychological-functional)
- Both have moderate conventionalization within respective communities but not across communities
- Positions are strongly tied to religious, political, and social identities

The gradient framework predicts this debate will not converge toward consensus through argument alone, because the semantic foundation operates at too low a conventionalization level. Resolution, if it occurs, must come through non-argumentative means: political compromise, institutional settlements, or cultural shifts that change conventionalization levels over generations.

Implications for Rational Discourse

The gradient framework has important implications for understanding rational discourse and disagreement:

Implication 1: Not all disagreements are resolvable through argument. When parties employ terms at low conventionalization levels, argument predictably fails to converge. This is not a failure of rationality but a structural feature of semantic organization. Recognizing this can reduce frustration and attribution of bad faith.

Implication 2: Semantic diagnosis is crucial. Before engaging in extended argument, parties should diagnose the conventionalization level of contested terms. If terms operate at sense or cultural connotation level, direct argument is unlikely to succeed. Metalinguistic negotiation or reframing may be more productive.

Implication 3: Productive disagreement requires shared semantic foundations. For argument to converge, parties need sufficient agreement on core meanings. This suggests the

value of building up from shared foundations rather than immediately engaging contested high-level terms.

Implication 4: Domain-specific variation. Some domains (mature sciences) operate at high conventionalization levels, enabling productive disagreement. Other domains (politics, ethics, philosophy) operate at lower levels, making convergence more difficult. This is not a defect but a consequence of what those domains study—contested normative concepts versus empirically tractable natural kinds.

Implication 5: Institutional role in conventionalization. Institutions can increase conventionalization through standardization, education, and enforcement. Scientific communities maintain high-conventionalization vocabularies through training, peer review, and correction of deviant usage. Political communities might benefit from similar mechanisms for stabilizing meanings of core political terms, though this faces obstacles from ideological diversity.

Predictions and Future Directions

The gradient framework generates numerous testable predictions and opens avenues for future research across multiple domains. This section outlines empirical predictions, identifies methodological approaches for testing them, and explores theoretical extensions.

Empirical Predictions About Semantic Structure

Prediction 1: Continuous distribution of agreement scores

If meaning is gradient rather than categorical, agreement scores should vary continuously rather than clustering at discrete points. When we measure $\alpha(f)$ across many features and words, the distribution should be continuous, not bimodal or multimodal.

Test: Conduct large-scale agreement studies measuring feature-word associations across diverse semantic domains. Participants rate association strength for hundreds of feature-word pairs. If the gradient view is correct, agreement scores should show continuous distribution. If

the categorical view is correct, scores should cluster at high values (lexical features) and low values (encyclopedic features) with a gap in between.

Prediction 2: Agreement-processing correlations

Features with higher conventionalization should show stronger and more consistent processing effects. Specifically:

- Higher $\alpha(f)$ should correlate with faster feature verification times
- Higher $\alpha(f)$ should correlate with stronger semantic priming effects
- Higher $\alpha(f)$ should correlate with larger N400 effects when violated
- These correlations should be continuous rather than showing discrete jumps

Test: Combine agreement measures with psycholinguistic experiments. For a set of words, measure both agreement scores (through norming studies) and processing measures (through verification tasks, priming studies, ERP experiments). Regression analyses should reveal continuous correlations between agreement and processing, controlling for confounds like frequency and imageability.

Prediction 3: Gradient acquisition trajectories

Language acquisition should show gradient patterns corresponding to conventionalization levels:

- Features with $\alpha \approx 1.0$ (reference) should be acquired earliest
- Features with $\alpha \approx x - n$ (denotation) should be acquired in early childhood
- Features with $\alpha \approx x - 2n$ (sense) should be acquired in middle childhood
- Features with $\alpha \approx x - 3n$ (cultural connotation) should be acquired in late childhood/adolescence
- Acquisition should be gradual rather than sudden for all levels

Test: Longitudinal developmental studies measuring children's understanding of different feature types across ages 2-15. Test both explicit knowledge (can the child articulate the feature?) and implicit knowledge (does the feature influence the child's usage?). The gradient view predicts continuous developmental trajectories with slopes corresponding to conventionalization levels.

Prediction 4: Cross-linguistic systematicity

For translation-equivalent words across languages, features should show correlated conventionalization levels. If feature f has $\alpha = 0.85$ for English word w_E and w_G is the German translation equivalent, then α for f in German should also be relatively high, even if not identical.

Test: Parallel agreement studies in typologically diverse languages. Measure $\alpha(f)$ for the same semantic features across translation equivalents. Calculate cross-linguistic correlations. The gradient view predicts moderate to strong correlations; pure conventionality would predict no correlation.

Prediction 5: Individual variation patterns

Individual speakers' agreement scores should pattern with demographic and experiential variables:

- More exposure to a register predicts higher agreement on register-specific features
- Professional training predicts higher agreement on domain-specific technical terms
- Cultural background predicts variation in cultural connotation levels
- Age correlates with semantic change: older speakers maintain older conventionalization patterns

**Test*:* Individual differences studies measuring agreement scores along with detailed demographic and experiential questionnaires. Multilevel models can test whether predicted variables explain individual variation in $\alpha(f)$ beyond baseline population levels.

Predictions About Entailment Behavior

Prediction 6: Cancellability-agreement correlation

The difficulty of canceling an entailment should correlate continuously with agreement level:

$$\text{Cancellability}(f) = k(1 - \alpha(f))$$

Features with $\alpha = 0.95$ should be nearly impossible to cancel; features with $\alpha = 0.60$ should be easily cancellable; features with $\alpha = 0.80$ should show intermediate cancellability.

Test: Acceptability judgment studies with cancellation frames: "X is a W but not F." Participants rate acceptability for various feature-word combinations with known agreement scores. Regression should reveal continuous negative correlation between $\alpha(f)$ and acceptability of cancellation.

Prediction 7: Compositional reinforcement and conflict

When words with similar weak entailments combine, their entailments should reinforce. When words with conflicting weak entailments combine, the result should be acceptable but marked:

$$P(f \mid [w_1 \ w_2]) = P(f \mid w_1) \cdot P(f \mid w_2) \cdot \lambda + (1 - \lambda)P(f \mid \text{context})$$

Test: Generate stimuli with reinforcing combinations (*utterly devastated*: both negative) and conflicting combinations (*utterly delightful*: conflicting). Measure acceptability ratings and processing times. Reinforcing combinations should be judged more natural and processed faster; conflicting combinations should be acceptable but show processing cost.

Prediction 8: Cross-construction projection

Weak entailments should project through compositional structure but be more easily cancelled than strong entailments:

- Negation cancels weak entailments more easily than strong ones
- Conditional antecedents weaken weak entailments more than strong ones
- Modal embedding allows weak entailments to vary across possible worlds

Test: Embed words with known weak entailments in various syntactic constructions (negation, conditionals, modals, questions). Measure whether the entailment projects, using both explicit judgments and implicit priming measures. Compare projection rates for strong, default, and weak entailments.

Predictions About Argument Dynamics

Prediction 9: Conventionalization level predicts argument outcomes

Arguments should show different dynamics depending on conventionalization level of contested terms:

- $\alpha \geq x - n$: Arguments converge through evidence
- $\alpha \approx x - 2n$: Arguments reach impasse without polarization
- $\alpha \approx x - 3n$: Arguments polarize, widening disagreement
- $\alpha < x - 3n$: No productive argument possible

Test: Controlled argument studies where pairs of participants discuss predetermined topics varying in conventionalization level of key terms. Measure pre-argument and post-argument belief distance. High-conventionalization topics should show convergence; moderate-conventionalization topics should show stability; low-conventionalization topics should show polarization.

Prediction 10: Metalinguistic negotiation success rates

Success rates for metalinguistic negotiation should correlate with initial conventionalization level. Negotiations starting near the sense level ($\alpha \approx 0.75$) have higher success rates than those starting at cultural connotation level ($\alpha \approx 0.60$).

Test: Experimental studies of metalinguistic negotiation. Pairs of participants are given terms at different conventionalization levels and asked to negotiate shared definitions. Success rates (proportion reaching mutual agreement) should correlate with initial α values.

Prediction 11: Discipline-level consensus variation

Academic disciplines should show different rates of consensus formation corresponding to average conventionalization levels of their core vocabularies:

- Natural sciences (high conventionalization): high consensus
- Social sciences (intermediate): moderate consensus
- Humanities (lower conventionalization): low consensus

Test: Bibliometric studies measuring consensus rates (citation patterns, co-citation networks, theoretical fragmentation) across disciplines. Combine with analysis of conventionalization levels of core terms in each discipline's vocabulary. Correlation should support the prediction.

Methodological Innovations

Several methodological approaches could advance testing of the gradient framework:

Method 1: Large-scale agreement norming

Develop comprehensive databases of agreement scores for thousands of feature-word pairs across multiple languages. This would provide empirical foundation for the framework analogous to what word frequency databases provide for usage-based linguistics.

Method 2: Computational agreement estimation

Train computational models to estimate $\alpha(f)$ from corpus distributions. If the gradient view is correct, distributional patterns should carry information about conventionalization levels. Models that successfully extract this information would provide evidence that agreement levels are encoded in usage patterns.

Method 3: Neural network probing for gradients

Use probing classifiers to test whether neural language models develop internal representations structured along conventionalization gradients. If the gradient is psychologically real, it should emerge in learned representations.

Method 4: Cross-linguistic parallel corpora

Develop parallel corpora with detailed semantic annotation for testing cross-linguistic predictions. These would enable testing whether conventionalization patterns are language-specific or show cross-linguistic universals.

Theoretical Extensions

Several theoretical extensions could deepen and broaden the framework:

Extension 1: Formal semantic implementation

Develop formal semantic systems that incorporate gradient conventionalization directly.

This might involve:

- Probabilistic type theories with gradient typing judgments
- Fuzzy logic frameworks with degrees of truth corresponding to agreement levels
- Dynamic semantics where conventionalization levels evolve through discourse
- Frame semantic systems with weighted feature structures

Extension 2: Integration with Construction Grammar

Examine how the gradient framework applies to constructional meanings beyond lexical meanings. Do constructions also show gradient conventionalization? How do lexical and constructional gradients interact in composition?

Extension 3: Social ontology of conventionalization

Investigate the social mechanisms that create and maintain conventionalization. What communication patterns, institutional structures, and social dynamics drive convergence toward higher agreement? How do power relations, authority structures, and group boundaries affect conventionalization processes?

Extension 4: Evolutionary semantics

Examine how conventionalization gradients emerge and evolve in populations through iterated learning and communication. Agent-based models could simulate the emergence of gradient semantic structure from individual learning and social interaction.

Extension 5: Neurocognitive implementation

Investigate neural mechanisms that could implement gradient conventionalization. How might neural architectures represent and process meanings at different agreement levels? What learning mechanisms could produce the observed gradient structure?

Limitations and Challenges

Several limitations and challenges require acknowledgment:

Challenge 1: Measurement precision

Measuring $\alpha(f)$ precisely is difficult. Different operationalizations (explicit judgments, corpus patterns, priming effects) may dissociate. The framework needs to specify how to handle such dissociations and what they mean theoretically.

Challenge 2: Compositionality

The compositional semantics of gradient meanings remains underspecified. How exactly do features at different conventionalization levels combine? The framework needs more precise composition principles, particularly for weak entailments.

Challenge 3: Cross-linguistic variation

The relationship between universal conceptual structure and language-specific conventionalization requires clarification. What aspects of the gradient are universal versus culturally contingent? How do we test this empirically?

Challenge 4: Normativity

The framework is primarily descriptive, but semantic theory often has normative dimensions. Should we encourage higher conventionalization for certain terms? How do we balance semantic precision against expressive flexibility?

Challenge 5: Computational complexity

Implementing the framework computationally may be expensive. Tracking conventionalization levels for all features of all words in real-time processing could be computationally prohibitive. The framework needs to address how cognitive systems manage this complexity.

Despite these challenges, the gradient framework represents significant theoretical progress in unifying disparate phenomena under a common explanatory structure. The next section integrates the semantic and epistemological applications, demonstrating the full scope of the unified framework.

Integration: A Unified Framework

The preceding sections have developed the gradient conventionalization framework through separate applications to lexical semantics and social epistemology. This section

integrates these applications, demonstrating how semantic prosody, connotative meaning, weak entailments, and argument dynamics all follow from a single unified theoretical structure.

The Core Architecture

At the foundation lies a single organizing principle: meaning is graded intersubjective agreement. Every semantic feature f associated with word w can be characterized by an agreement score $\alpha(f) \in [0,1]$ representing the proportion of speakers who share that association. This simple quantitative measure organizes the entire semantic landscape.

From this foundation, several theoretical structures emerge:

Level 1: The Conventionalization Gradient

Features distribute continuously along the gradient according to their agreement scores:

Reference $\xrightarrow{\alpha \approx 1.0}$ Denotation $\xrightarrow{\alpha \approx 0.85}$ Sense $\xrightarrow{\alpha \approx 0.75}$ Cultural Connotation $\xrightarrow{\alpha \approx 0.60}$ Private Connotation

This gradient replaces the categorical lexical-encyclopedia boundary with a continuous space of conventionalization. Features previously forced into one category or the other can be positioned precisely according to their empirically measured agreement levels.

Level 2: The Entailment Typology

The conventionalization gradient generates a corresponding entailment typology:

Strong Entailments $\xrightarrow{\alpha \geq 0.85}$ Default Entailments $\xrightarrow{\alpha \approx 0.75}$ Weak Entailments $\xrightarrow{\alpha \approx 0.60}$ No Entailments

This typology provides the tools for understanding how meanings at different conventionalization levels constrain inference. Strong entailments function as necessary inferences, default entailments as overridable expectations, and weak entailments as probabilistic tendencies.

Level 3: The Argument Dynamics

The same conventionalization levels that determine entailment strength also determine argument dynamics:

$$\text{Convergence} \xleftarrow{\alpha \geq 0.85} \text{Impasse} \xleftarrow{\alpha \approx 0.75} \text{Polarization} \xleftarrow{\alpha \approx 0.60} \text{Impossibility}$$

Arguments employing terms at high conventionalization levels converge through evidence and reasoning. Arguments at intermediate levels reach unproductive impasses. Arguments at low levels actively polarize, widening rather than narrowing disagreements.

These three levels—gradient, entailments, arguments—constitute a unified architecture. Each level follows from the previous through lawful principles:

$$\alpha(f) \Rightarrow \text{Entailment Type}(f) \Rightarrow \text{Argument Dynamic}(f)$$

The agreement score determines the entailment type, which determines the argument dynamic. The entire system reduces to a single parameter: intersubjective agreement.

Unifying Semantic Phenomena

The framework unifies several previously disparate semantic phenomena under common principles:

Phenomenon 1: Semantic Prosody

Semantic prosody, previously treated as *sui generis* or dismissed as statistical artifact, emerges as cultural connotation generating weak entailments. Words like *cause*, *regime*, and *utterly* encode semantic profiles with features at cultural connotation level ($\alpha \approx 0.60$). These features create probabilistic preferences for particular contexts without absolute constraints.

The distributional patterns observed in corpora result from weak entailments interacting with discourse pragmatics. *Cause* appears predominantly with negative outcomes not because it has absorbed negative meaning but because its agentive-forceful profile (a feature with $\alpha \approx 0.65$) aligns naturally with contexts where forceful imposition is appropriate, which correlates with negative outcomes in discourse.

Counterexamples like "cause celebration" remain acceptable because the entailment is weak ($\alpha \approx 0.60$ rather than $\alpha \geq 0.90$). The usage is somewhat marked, creating rhetorical effects, but not contradictory. This explains the puzzle of systematicity without consciousness: the features have sufficient conventionalization to influence processing systematically but insufficient conventionalization for reliable metalinguistic access.

Phenomenon 2: Connotation and Denotation

The traditional distinction between denotation (core definitional meaning) and connotation (peripheral associations) dissolves into a gradient. What we call denotation comprises features with very high agreement ($\alpha \geq 0.85$); what we call connotation comprises features with moderate agreement ($\alpha \approx 0.50 - 0.70$). The distinction is quantitative rather than qualitative.

This resolves the persistent instability of the denotation-connotation boundary. Different theorists draw the line at different agreement thresholds, producing apparently conflicting analyses of the same phenomena. Under the gradient view, there is no single correct threshold—only points along a continuous space that different frameworks sample differently.

Moreover, the framework explains why cultural connotation shows systematicity (sufficient agreement to generate corpus patterns) while private connotation does not (insufficient agreement to produce systematic effects). The threshold appears around $\alpha \approx 0.50$: above this, patterns emerge; below it, idiosyncrasy dominates.

Phenomenon 3: Vagueness and Borderline Cases

Vagueness arises from applying features with high but not perfect agreement ($\alpha \approx 0.80 - 0.90$) to borderline cases where the feature applicability is itself uncertain. Consider *bald*. Most speakers agree that complete absence of head hair constitutes baldness ($\alpha \approx 0.95$). But for intermediate cases—receding hairlines, thin coverage—agreement decreases ($\alpha \approx 0.60 - 0.75$), creating vagueness.

The framework distinguishes two types of indeterminacy:

***Type 1 Vagueness*:** High agreement on feature ($\alpha \geq 0.85$) combined with uncertain feature applicability. Is this man bald? We all agree bald means absence of hair, but we're uncertain whether this hair count qualifies as absence.

***Type 2 Vagueness*:** Moderate agreement on feature ($\alpha \approx 0.70$) producing disagreement about application. Is tomato a fruit? We disagree about which features define fruit-hood, producing divergent judgments.

This distinction parallels the linguistic/metalinguistic ambiguity distinction in vagueness research but grounds it in measurable agreement levels rather than theoretical stipulation.

Phenomenon 4: Polysemy and Sense Relations

Polysemy—the phenomenon of words having multiple related meanings—can be understood through the gradient framework. Different senses of a word correspond to different regions of the conventionalization gradient being more or less activated in different contexts.

Consider *bank* (financial institution vs. river edge). These are discrete senses with high agreement within each sense ($\alpha \geq 0.90$) but no agreement across senses ($\alpha \approx 0$). This is categorical polysemy. But many cases show gradient polysemy, where senses blur into each other.

Consider *run*: to move quickly by foot, to operate (run a business), to flow (water runs), to extend (road runs north), to function (car runs well). These senses show gradient relationships. Some features are shared across multiple senses (dynamic process, continuation over time) with high agreement; others are sense-specific with moderate agreement. The gradient framework captures this graded relatedness through continuous variation in agreement scores across the feature space.

Phenomenon 5: Prototype Effects

Prototype effects—the finding that category members vary in typicality—follow naturally from the gradient framework. More typical members satisfy more features with higher agreement scores; less typical members satisfy fewer features or features with lower agreement.

For **bird**, a robin satisfies many high-agreement features: flies ($\alpha \approx 0.85$), has wings ($\alpha \approx 0.95$), lays eggs ($\alpha \approx 0.90$), sings ($\alpha \approx 0.75$). A penguin satisfies the highest-agreement features (has wings, lays eggs) but not moderate-agreement features (flies, sings). Typicality correlates with weighted satisfaction of conventionalized features:

$$\text{Typicality}(x, C) = \sum_{f \in \text{Features}(C)} \alpha(f) \cdot \mathbb{1}[x \text{ has } f]$$

where $\mathbb{1}[x \text{ has } f]$ indicates whether exemplar x possesses feature f . This formula predicts typicality ratings from agreement scores and feature presence, unifying prototype theory with the gradient framework.

Unifying Epistemological Phenomena

The framework equally unifies diverse phenomena in social epistemology and argumentation:

Phenomenon 6: Scientific Consensus Formation

Scientific communities converge toward consensus because they operate at high conventionalization levels. Technical terms in mature sciences have agreement scores $\alpha \geq 0.85$, enabling:

1. Shared inference rules: When $p \models q$ is strongly entailed, all practitioners accept the inference
2. Cumulative knowledge: Established results serve as foundations because all accept the underlying meanings
3. Resolvable disputes: Disagreements about cases reduce to empirical questions about feature satisfaction

The framework predicts variation across sciences. Fields with higher-conventionalization vocabularies (physics, chemistry) should show faster consensus formation than fields with lower-conventionalization vocabularies (sociology, literary criticism). Bibliometric data support this prediction (Cole, 1983).

Phenomenon 7: Philosophical Persistence

Philosophical disputes persist because philosophy operates at sense level ($\alpha \approx 0.70 - 0.85$). Core philosophical terms like *knowledge*, *justice*, *consciousness*, *free will* have high denotation-level agreement but substantial sense-level variation.

Different philosophical schools organize these concepts around different features, each with moderate conventionalization within schools but lower conventionalization across schools. This produces the characteristic pattern of persistent pluralism: multiple theories each capturing important features of the concept, none achieving dominance.

The framework predicts that philosophy will not converge through argument alone, because the semantic foundation is insufficiently shared. Progress requires either:

- Increasing conventionalization through sustained community effort
- Fragmenting into subfields with higher internal conventionalization
- Accepting pluralism as reflecting genuine variation in semantic organization

Phenomenon 8: Political Polarization

Political polarization results from arguments employing terms at cultural connotation level ($\alpha \approx 0.50 - 0.70$). Terms like *freedom*, *justice*, *rights*, *equality* show moderate population-level agreement but substantial variation across political subgroups.

Arguments over such terms predictably polarize through the mechanisms identified earlier: confirmation bias amplification, motivated reasoning about interlocutors, identity reinforcement, and semantic divergence. The framework explains why polarization has increased in recent decades: political discourse has shifted toward arguing about contested high-level abstractions rather than concrete policy details.

The framework suggests interventions: reframing political arguments to employ higher-conventionalization terms, building up from shared foundations, or acknowledging semantic differences and negotiating rather than arguing.

Phenomenon 9: Metalinguistic Negotiation Patterns

The success or failure of metalinguistic negotiation depends on initial conventionalization levels. Negotiation succeeds when:

- Starting agreement is relatively high ($\alpha \geq 0.70$)
- Parties share meta-principles for adjudicating between conceptions
- Institutional mechanisms can reinforce negotiated meanings

Negotiation fails when:

- Starting agreement is low ($\alpha < 0.60$)
- Feature sets are genuinely different across parties

- Identity commitments are tied to particular conceptions

This explains patterns across domains. Legal negotiation often succeeds because legal communities have institutional mechanisms for enforcing conventionalized meanings. Political negotiation often fails because political communities lack such mechanisms and feature strong identity commitments.

Mathematical Unification

The framework can be unified mathematically through a compositional model that derives all phenomena from the agreement function $\alpha: \mathcal{F} \rightarrow [0,1]$.

Definition 1: Semantic Content

The semantic content of word w is a weighted distribution over features:

$$\text{Sem}(w) = \{(f_1, \alpha(f_1)), (f_2, \alpha(f_2)), \dots, (f_n, \alpha(f_n))\}$$

Definition 2: Entailment Strength

For feature f and proposition $p(w)$, the entailment strength is:

$$\text{Strength}(p(w) \vdash f) = \begin{cases} \text{strong} & \text{if } \alpha(f) \geq x - n \\ \text{default} & \text{if } x - 2n \leq \alpha(f) < x - n \\ \text{weak} & \text{if } x - 3n \leq \alpha(f) < x - 2n \\ \text{none} & \text{if } \alpha(f) < x - 3n \end{cases}$$

Definition 3: Cancellability

The cancellability of feature f is inversely proportional to agreement:

$$\text{Cancel}(f) = k \cdot (1 - \alpha(f))$$

where k is a scaling constant. This predicts continuous variation in ease of cancellation.

Definition 4: Argument Convergence

For arguers A and B discussing proposition $p(w)$ involving feature f , the expected change in belief distance is:

$$\mathbb{E}[\Delta(\mathcal{B}_A, \mathcal{B}_B)] = \begin{cases} < 0 & \text{if } \alpha(f) \geq x - n \\ \approx 0 & \text{if } x - 2n \leq \alpha(f) < x - n \\ > 0 & \text{if } \alpha(f) < x - 2n \end{cases}$$

Negative values indicate convergence, zero indicates stability, positive values indicate polarization.

Theorem 1: Semantic-Epistemological Correspondence

For any feature f , entailment strength determines argument dynamics:

$$\text{Strength}(p(w) \vdash f) = \text{strong} \Rightarrow \mathbb{E}[\Delta(\mathcal{B}_A, \mathcal{B}_B)] < 0$$

$$\text{Strength}(p(w) \vdash f) = \text{default} \Rightarrow \mathbb{E}[\Delta(\mathcal{B}_A, \mathcal{B}_B)] \approx 0$$

$$\text{Strength}(p(w) \vdash f) = \text{weak} \Rightarrow \mathbb{E}[\Delta(\mathcal{B}_A, \mathcal{B}_B)] > 0$$

This theorem unifies semantics and epistemology: the semantic property (entailment strength) determines the epistemological property (argument dynamics) through the common foundation (agreement level).

Proof Sketch: Strong entailments arise when $\alpha(f) \geq x - n$, meaning near-universal agreement. When both arguers accept the entailment, they share inference rules and can rationally persuade each other, producing convergence: $\Delta < 0$. Default entailments arise when $x - 2n \leq \alpha(f) < x - n$, meaning high but not universal agreement. Arguers may organize semantic fields differently, leading to impasse without polarization: $\Delta \approx 0$. Weak entailments arise when $x - 3n \leq \alpha(f) < x - 2n$, meaning moderate agreement. Arguers likely have different associations, leading to talking-past-each-other and polarization: $\Delta > 0$. $\square$

This mathematical unification demonstrates that the entire framework—from semantic prosody to political polarization—follows from a single foundational principle: meaning is graded intersubjective agreement.

Explanatory Breadth and Parsimony

The unified framework achieves remarkable explanatory breadth from minimal theoretical apparatus. Starting only from:

1. The agreement function $\alpha: \mathcal{F} \rightarrow [0,1]$
2. The principle that entailment strength correlates with agreement
3. The principle that argument dynamics correlate with entailment strength

We derive explanations for:

- Why the lexical-encyclopedic boundary is unstable (it's a gradient, not a boundary)
- Why semantic prosody shows systematicity without consciousness (moderate conventionalization influences processing without metalinguistic access)
- Why counterexamples to prosodic patterns are acceptable (weak entailments are easily overridden)
- Why connotation has been undertheorized (cultural and private connotation were conflated)
- Why scientific communities converge (high-conventionalization vocabularies)
- Why philosophical communities don't (sense-level operation)
- Why political arguments polarize (cultural connotation-level terms)
- Why some metalinguistic negotiations succeed and others fail (depends on initial agreement)

This breadth from parsimony represents theoretical progress. Rather than requiring separate theories for each phenomenon, the gradient framework provides a unified account.

Advantages Over Alternative Frameworks

The gradient framework has several advantages over competing approaches:

Advantage 1: Empirical Grounding

Unlike categorical frameworks that stipulate boundaries, the gradient framework grounds distinctions in measurable quantities: agreement scores. These can be operationalized through multiple converging methods (explicit judgments, corpus patterns, processing measures, computational models), providing empirical foundation for theoretical claims.

Advantage 2: Predictive Precision

The framework generates precise quantitative predictions. Rather than qualitative predictions ("connotation is variable"), it predicts specific correlations (cancellability varies continuously with $1 - \alpha(f)$) and thresholds (polarization occurs when $\alpha < x - 2n$). These predictions are falsifiable through empirical testing.

Advantage 3: Cross-Domain Integration

The framework unifies semantics and epistemology, showing how the same structural principles govern both lexical meaning and argumentation. This cross-domain integration is not ad hoc but follows from the mathematics: agreement levels determine both entailment strength and argument dynamics.

Advantage 4: Gradualist Ontology

By treating meaning as gradient, the framework avoids problems that plague categorical approaches. There are no boundary disputes because there are no boundaries—only points along a continuum. This aligns with converging evidence from psycholinguistics, neuroscience, and computational modeling that semantic knowledge is graded rather than categorical.

Advantage 5: Computational Tractability

The framework is computationally implementable. Neural language models can estimate agreement scores from distributional evidence, represent gradient semantic structures, and simulate argument dynamics. This implementability provides both a test of the framework and a bridge to applications in natural language processing.

Remaining Integration Challenges

Despite its explanatory power, several integration challenges remain:

Challenge 1: Compositional Semantics

While the framework specifies how individual features behave at different conventionalization levels, the compositional principles for combining features across conventionalization levels need further development. How exactly do strong entailments compose with weak ones? Does composition preserve, amplify, or attenuate conventionalization levels?

A preliminary proposal: composition preserves the minimum conventionalization level of composed elements. If w_1 has feature f_1 with $\alpha(f_1) = 0.85$ and w_2 has feature f_2 with $\alpha(f_2) = 0.65$, then $[w_1 w_2]$ has the composite feature with $\alpha \approx \min(0.85, 0.65) = 0.65$. This predicts that weak entailments in composition "dilute" the overall conventionalization, making composed expressions more variable than their parts.

But this is only a first approximation. Different composition types (modification, complementation, coordination) may show different interaction patterns. Detailed compositional theory remains a priority for future work.

Challenge 2: Dynamic Update

The framework treats $\alpha(f)$ as static, but agreement levels change over time through semantic change, dialect formation, and individual learning. A fully dynamic framework would specify:

$$\frac{d\alpha(f, t)}{dt} = g(\text{usage patterns, social dynamics, individual learning})$$

Modeling these dynamics would enable predictions about semantic change trajectories, dialect divergence, and acquisition timelines. However, specifying the function g requires integrating insights from historical linguistics, sociolinguistics, and developmental psycholinguistics—a substantial undertaking.

Challenge 3: Cross-Linguistic Variation

While the framework makes predictions about cross-linguistic correlations in conventionalization levels, it does not yet specify which aspects of semantic structure are universal versus language-specific. Do all languages show the same five-level gradient, or do the number and positions of levels vary? Do typological differences (isolating vs. agglutinative vs. fusional) affect conventionalization patterns?

Answering these questions requires extensive cross-linguistic research grounded in the gradient framework, comparing agreement scores across diverse language families.

Challenge 4: Neural Implementation

The framework needs to specify how gradient semantic structures are implemented neurally. What neural coding schemes represent agreement levels? How do processing networks distinguish strong from weak entailments? What learning mechanisms produce gradient conventionalization from linguistic input?

Recent work on semantic gradients in neural networks provides starting points (Grand et al., 2022), but full neural grounding remains an open challenge.

Despite these challenges, the integrated framework represents significant theoretical progress. By unifying diverse phenomena under common principles grounded in measurable quantities, it provides a foundation for future empirical and theoretical work across semantics, epistemology, and related fields.

Conclusion

This paper has proposed a fundamental reconceptualization of the relationship between lexical meaning and world knowledge. Rather than maintaining a categorical distinction between types of knowledge, I have argued that all meaning exists on a continuous gradient of conventionalization, operationalized as degrees of intersubjective agreement among speakers. What we call "lexical" meaning simply represents the highest levels of conventionalization, while what we call "encyclopedic" knowledge occupies lower positions on the same gradient.

This reconceptualization resolves several longstanding problems in semantic theory. It explains why the lexical-encyclopedic boundary has proven persistently unstable: theorists have been attempting to draw categorical lines through continuous space. It provides a natural account of intermediate cases that trouble categorical approaches. And it aligns with converging evidence from the arbitrariness of linguistic signs, language acquisition patterns, computational modeling success, cross-linguistic variation, and psycholinguistic processing studies.

The gradient framework required a parallel reconceptualization of entailments. Rather than treating entailment as a categorical relation, I developed a richer typology recognizing strong entailments (necessary inferences from features with $\alpha \geq 0.85$), default entailments (overridable inferences from features with $\alpha \approx 0.75$), and weak entailments (probabilistic inferences from features with $\alpha \approx 0.60$). This typology provides the theoretical tools needed for

understanding cultural connotation—the crucial middle ground between strong semantic content and idiosyncratic association.

The key diagnostic error in prior work has been conflating two distinct phenomena under "connotation": cultural connotation (systematic, community-level patterns with moderate agreement) and private connotation (idiosyncratic associations with minimal agreement). By failing to distinguish these, semantic theory treated all connotation as private, concluding it lies outside semantics proper. This conclusion is premature. Cultural connotation exhibits sufficient systematicity to warrant formal treatment and can be understood as generating weak entailments at moderate conventionalization levels.

Application to semantic prosody demonstrated the empirical utility of this framework. Prosodic patterns, which have resisted satisfactory theoretical treatment in traditional frameworks, emerge naturally as cultural connotations generating weak entailments. The framework explains prosody's apparently paradoxical properties: systematic yet intuition-resistant, statistical yet structured, meaningful yet non-truth-conditional. These properties all follow from intermediate conventionalization levels and the weak entailmental patterns they generate.

Beyond lexical semantics, the framework provides theoretical tools for understanding social epistemology and argument dynamics. The same gradient of conventionalization that distinguishes types of lexical meaning also predicts patterns in argumentative discourse. Arguments employing high-conventionalization terms converge through evidence and reasoning. Arguments employing intermediate-conventionalization terms reach unproductive impasses. Arguments employing low-conventionalization terms actively polarize, widening rather than narrowing disagreements.

This connection explains several puzzling asymmetries: why scientific communities converge while philosophical communities do not; why political discourse has become increasingly polarized; why some metalinguistic negotiations succeed while others fail. The semantic structure of the terms employed in arguments constrains the dynamics of belief revision and consensus formation through lawful principles that can be formally specified and empirically tested.

The theoretical implications extend broadly. For semantic theory, the framework suggests the need for representational systems that can capture graded distinctions in conventionalization and entailment strength. Classical truth-conditional semantics, with its categorical approach, must be enriched to handle the middle ranges of the gradient. For acquisition research, the framework predicts correlation between conventionalization levels and acquisition timelines. For computational modeling, it explains why distributional learning succeeds at acquiring both core meanings and cultural connotations—the patterns encode gradient semantic structure recoverable through statistical learning. For cross-linguistic semantics, it provides tools for understanding systematic variation in how languages conventionalize semantic features. For diachronic semantics, it offers mechanisms for understanding semantic change as movement along the conventionalization gradient.

Much work remains. The compositional semantics of weak entailments requires further specification. Empirical validation through systematic psycholinguistic experimentation is needed to test predicted correlations between conventionalization levels, agreement scores, and processing measures. Cross-linguistic investigation could reveal whether the gradient structure is universal or subject to typological variation. Exploration of neural implementation could provide insights into cognitive underpinnings of semantic representation. Development of formal

frameworks for representing and composing weak entailments remains a significant theoretical challenge.

Nevertheless, the fundamental insight seems robust: meaning is not categorically partitioned into distinct types but rather exists on a continuous gradient of conventionalization. Recognizing this gradient structure, and the system of weak entailments it generates, provides theoretical tools for engaging with the full richness of systematic meaning phenomena while avoiding the pitfalls of attempting to maintain unstable categorical boundaries.

The unified framework developed here demonstrates that the persistent puzzle of how words mean and how meanings constrain usage may find resolution not in discovering the "true" boundary between types of meaning but in recognizing that no such boundary exists—only points along a continuous landscape of conventionalization. Similarly, the epistemological puzzles about when arguments converge and when they polarize find resolution not in methodological differences but in the semantic structure of the terms employed. The gradient of conventionalization shapes both lexical meaning and social epistemology through common principles grounded in a single measurable quantity: intersubjective agreement.

This reconceptualization opens new avenues for semantic theory while grounding its claims in empirically observable patterns of agreement, distributional consistency, and processing effects. By providing a framework that unifies lexical semantics, semantic prosody, connotative meaning, compositional semantics, language acquisition, computational modeling, cross-linguistic variation, semantic change, social epistemology, argument dynamics, consensus formation, and metalinguistic negotiation under common principles, the gradient conventionalization framework represents a significant step toward a comprehensive theory of meaning that is both empirically grounded and explanatorily powerful.

The theoretical payoff extends beyond resolving specific empirical puzzles. By providing tools for formalizing graded meaning distinctions and weak entailments, this framework enables semantic theory to engage with a wider range of systematic linguistic phenomena while maintaining theoretical rigor. Moreover, it offers a more psychologically and computationally plausible model of how meaning is represented and processed—one that aligns with contemporary evidence from acquisition studies, neurolinguistics, machine learning, and usage-based linguistics. The path forward requires continued dialogue among corpus linguistics, formal semantics, psycholinguistics, computational approaches, and philosophical analysis—each contributing essential insights to our understanding of how meaning shapes, and is shaped by, the statistical fabric of language use and the dynamics of social communication.

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